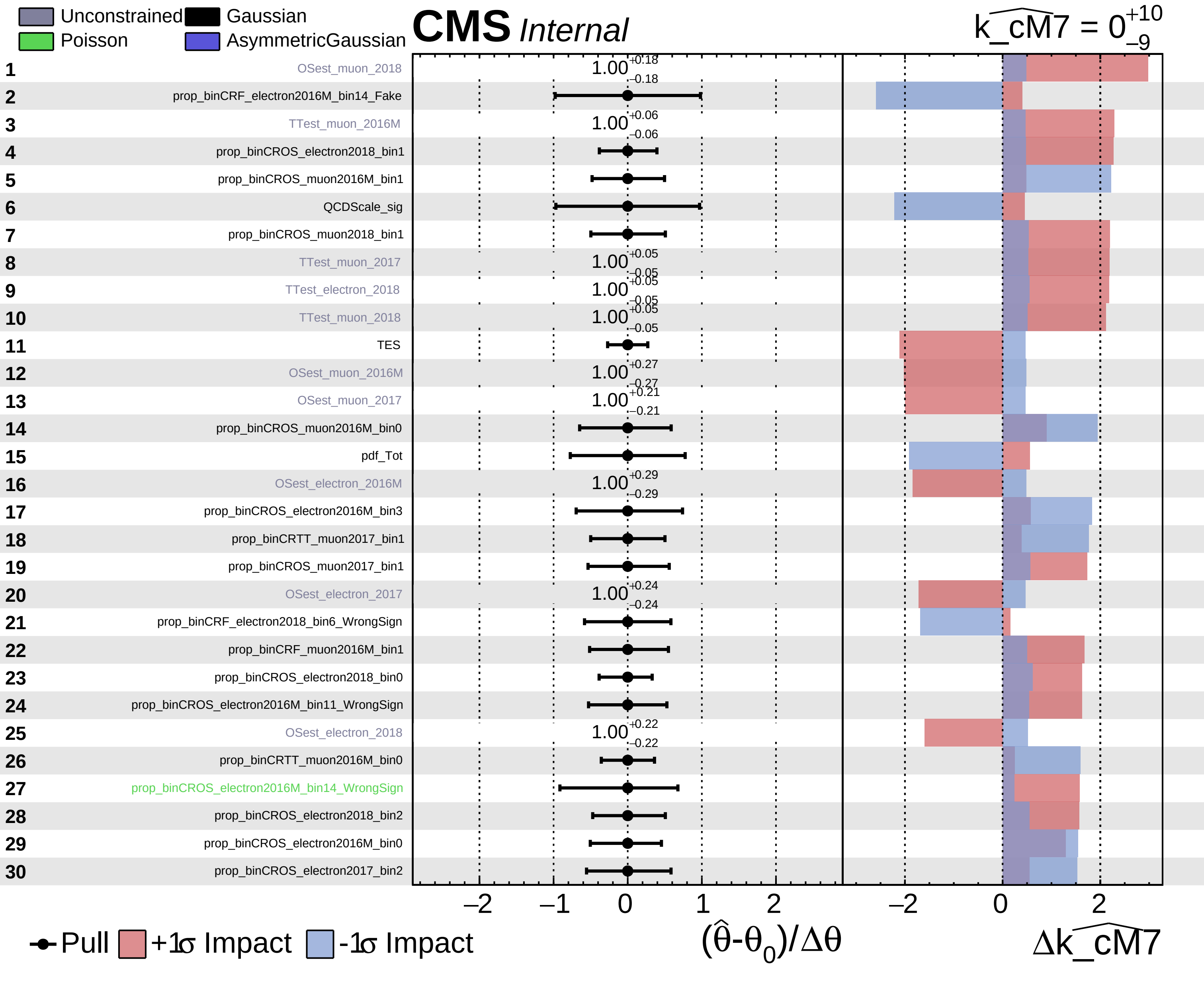
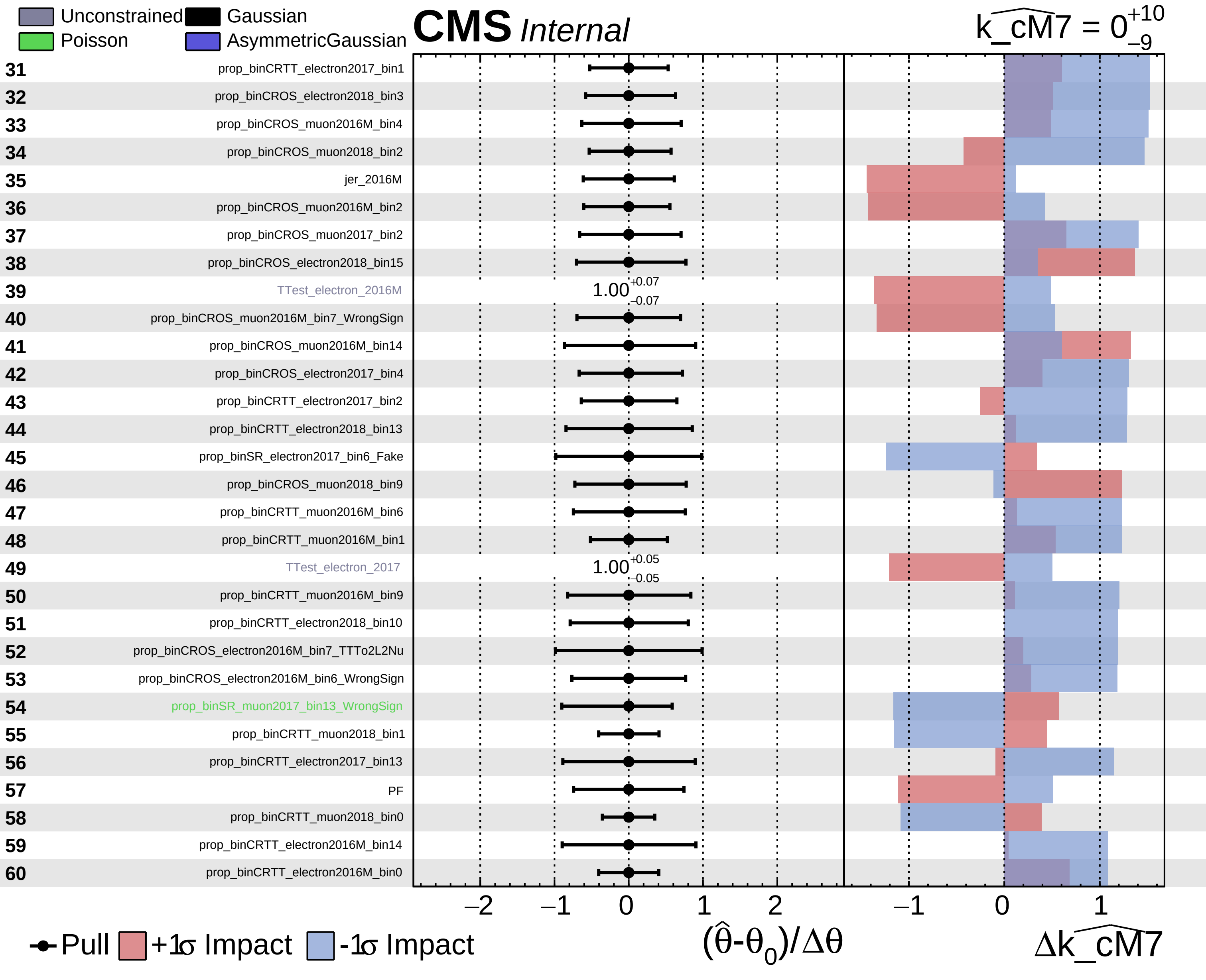
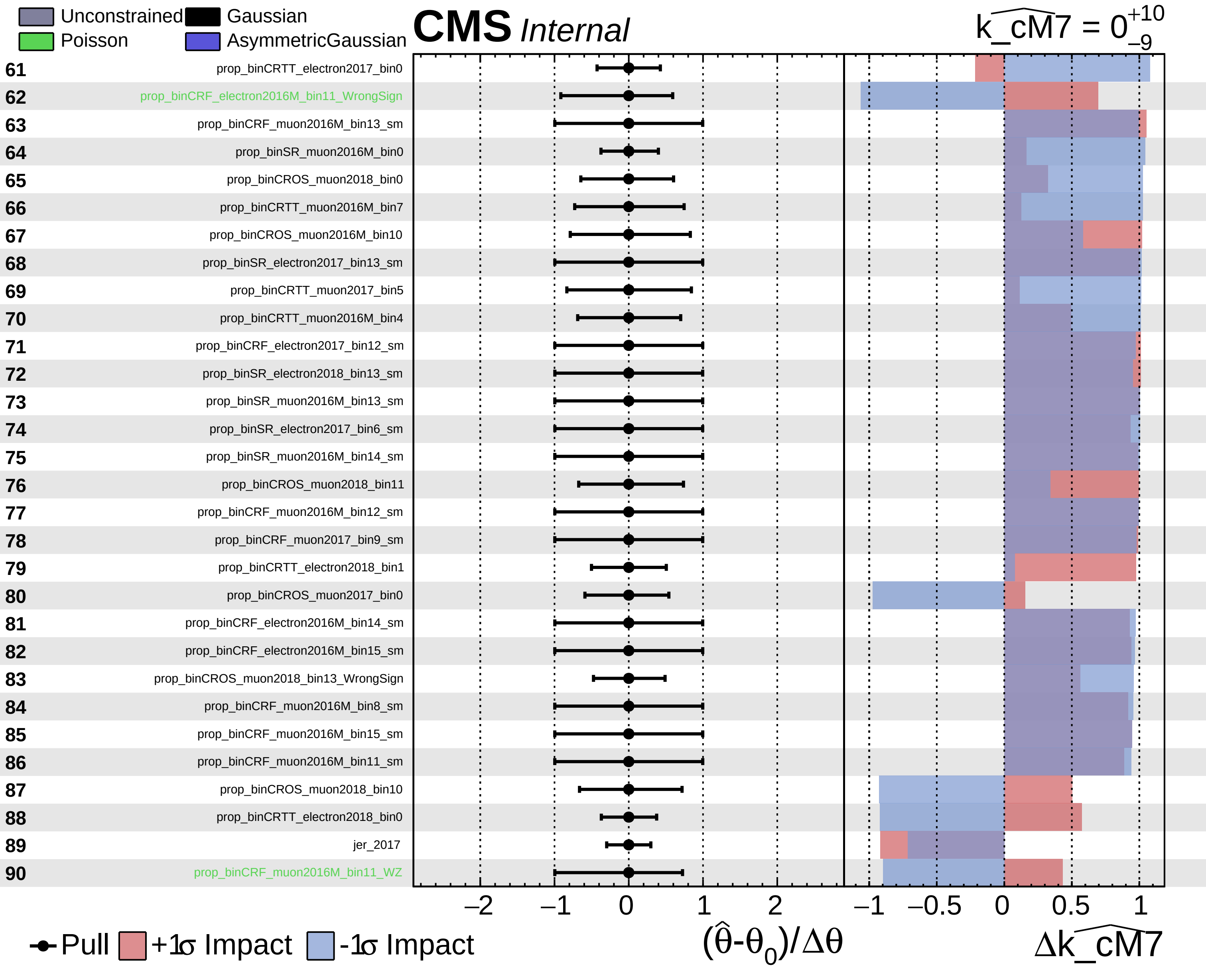
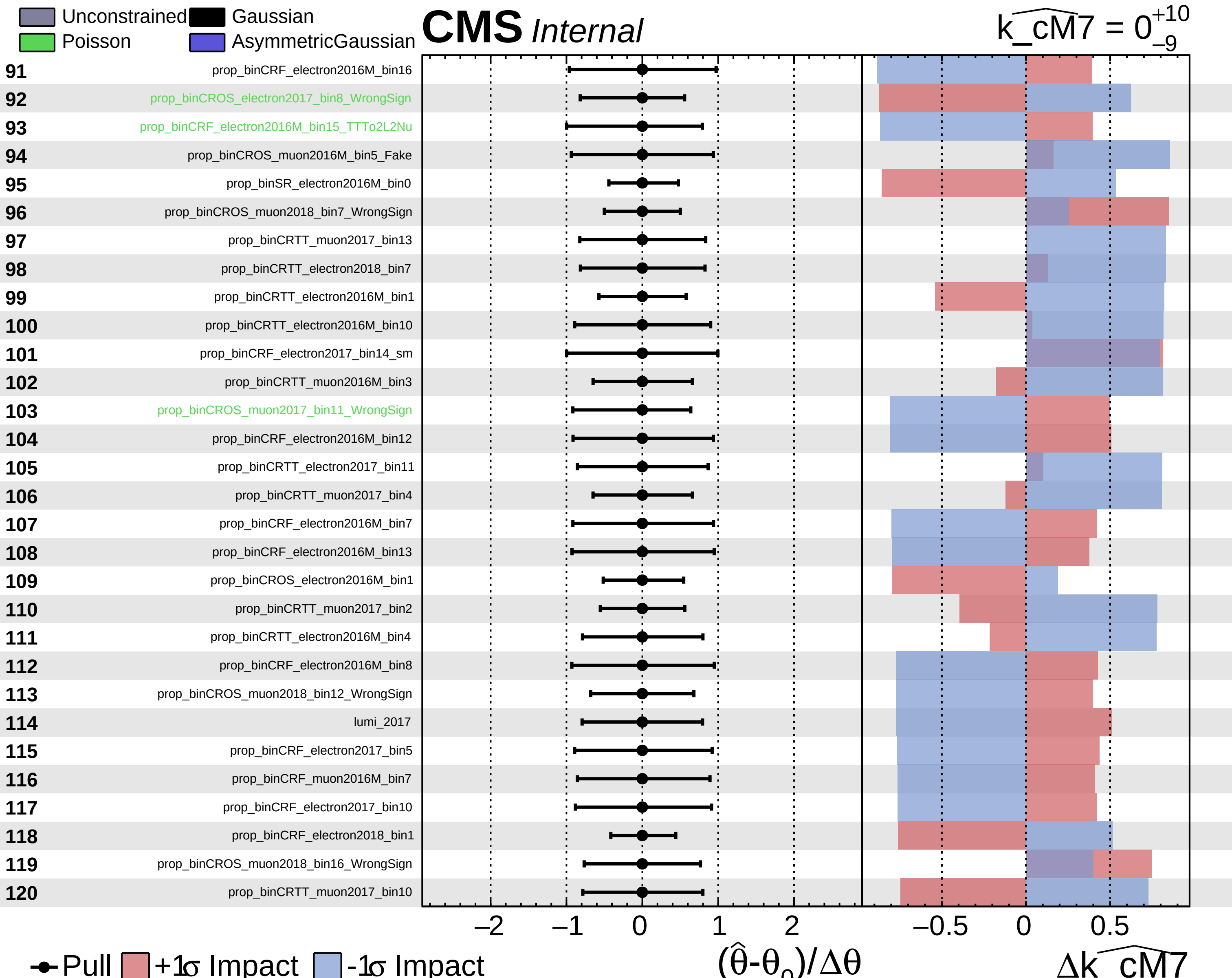


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
1 OSest_muon_2018 $1.00^{+0.18}_{-0.18}$
2 prop_binCRF_electron2016M_bin14_Fake
3 TTest_muon_2016M $1.00^{+0.06}_{-0.06}$
4 prop_binCROS_electron2018_bin1
5 prop_binCROS_muon2016M_bin1
6 QCDScale_sig
7 prop_binCROS_muon2018_bin1
8 TTest_muon_2017 $1.00^{+0.05}_{-0.05}$
9 TTest_electron_2018 $1.00^{+0.05}_{-0.05}$
10 TTest_muon_2018 $1.00^{+0.05}_{-0.05}$
11 TES
12 OSest_muon_2016M $1.00^{+0.27}_{-0.27}$
13 OSest_muon_2017 $1.00^{+0.21}_{-0.21}$
14 prop_binCROS_muon2016M_bin0
15 pdf_Tot
16 OSest_electron_2016M $1.00^{+0.29}_{-0.29}$
17 prop_binCROS_electron2016M_bin3
18 prop_binCRTT_muon2017_bin1
19 prop_binCROS_muon2017_bin1
20 OSest_electron_2017 $1.00^{+0.24}_{-0.24}$
21 prop_binCRF_electron2018_bin6_WrongSign
22 prop_binCRF_muon2016M_bin1
23 prop_binCROS_electron2018_bin0
24 prop_binCROS_electron2016M_bin11_WrongSign
25 OSest_electron_2018 $1.00^{+0.22}_{-0.22}$
26 prop_binCRTT_muon2016M_bin0
27 prop_binCROS_electron2016M_bin14_WrongSign
28 prop_binCROS_electron2018_bin2
29 prop_binCROS_electron2016M_bin0
30 prop_binCROS_electron2017_bin2
−2 −1 0 1 2
−2 0 2
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$

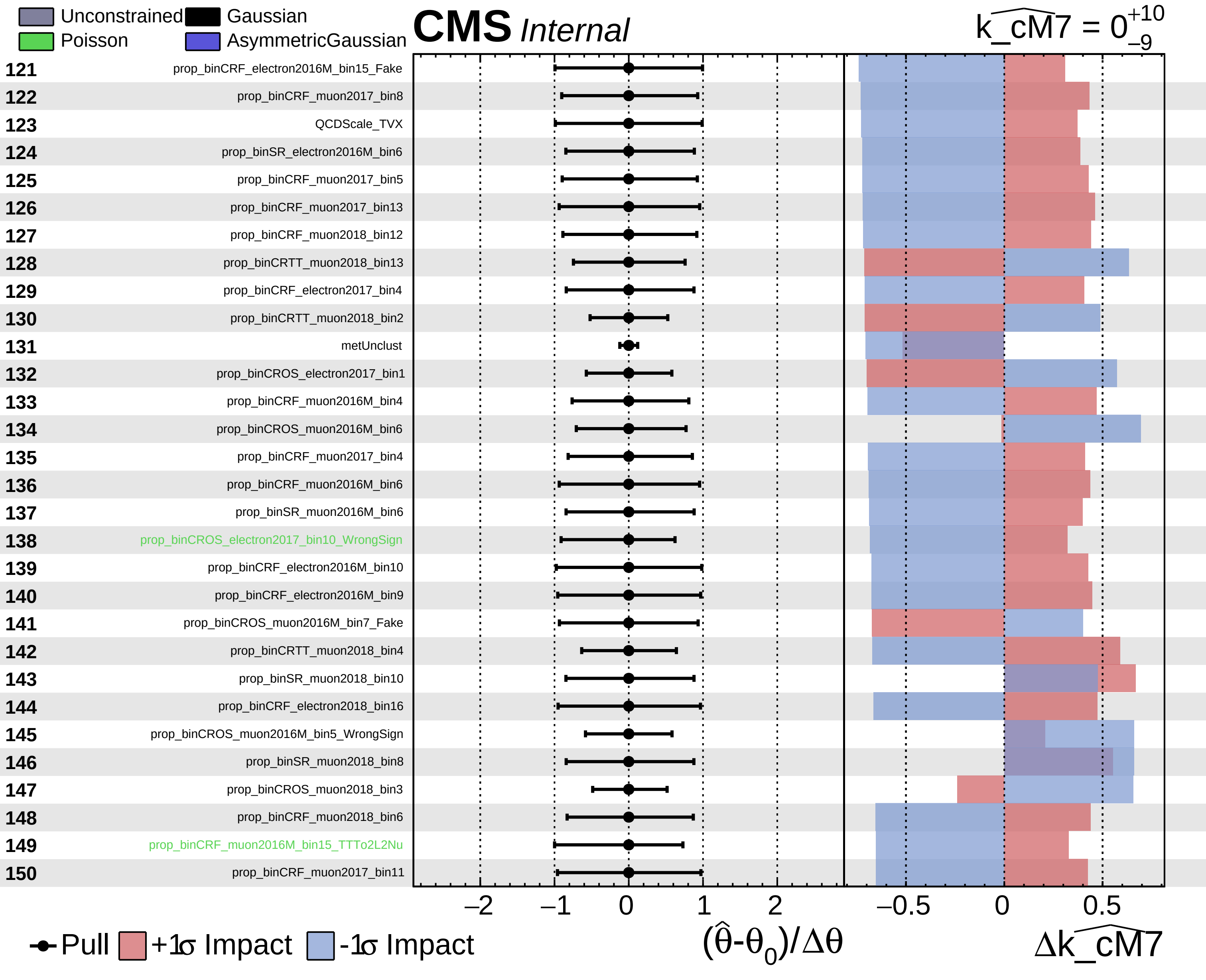


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
31 prop_binCRTT_electron2017_bin1
32 prop_binCROS_electron2018_bin3
33 prop_binCROS_muon2016M_bin4
34 prop_binCROS_muon2018_bin2
35 jer_2016M
36 prop_binCROS_muon2016M_bin2
37 prop_binCROS_muon2017_bin2
38 prop_binCROS_electron2018_bin15
39 TTest_electron_2016M
$1.00^{+0.07}_{-0.07}$
40 prop_binCROS_muon2016M_bin7_WrongSign
41 prop_binCROS_muon2016M_bin14
42 prop_binCROS_electron2017_bin4
43 prop_binCRTT_electron2017_bin2
44 prop_binCRTT_electron2018_bin13
45 prop_binSR_electron2017_bin6_Fake
46 prop_binCROS_muon2018_bin9
47 prop_binCRTT_muon2016M_bin6
48 prop_binCRTT_muon2016M_bin1
49 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
50 prop_binCRTT_muon2016M_bin9
51 prop_binCRTT_electron2018_bin10
52 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
53 prop_binCROS_electron2016M_bin6_WrongSign
54 prop_binSR_muon2017_bin13_WrongSign
55 prop_binCRTT_muon2018_bin1
56 prop_binCRTT_electron2017_bin13
57 PF
58 prop_binCRTT_muon2018_bin0
59 prop_binCRTT_electron2016M_bin14
60 prop_binCRTT_electron2016M_bin0
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$

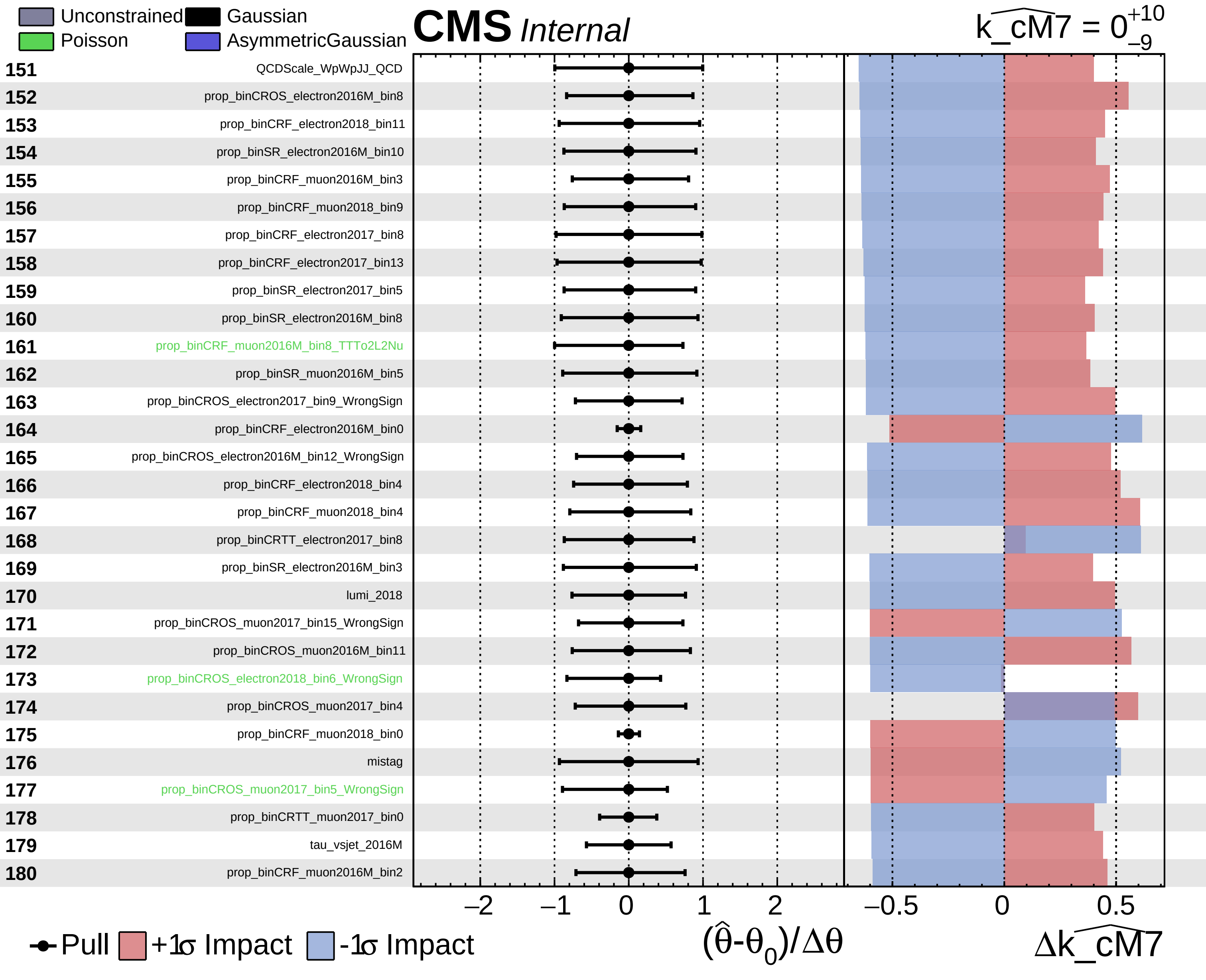

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
61 prop_binCRTT_electron2017_bin0
62 prop_binCRF_electron2016M_bin11_WrongSign
63 prop_binCRF_muon2016M_bin13_sm
64 prop_binSR_muon2016M_bin0
65 prop_binCROS_muon2018_bin0
66 prop_binCRTT_muon2016M_bin7
67 prop_binCROS_muon2016M_bin10
68 prop_binSR_electron2017_bin13_sm
69 prop_binCRTT_muon2017_bin5
70 prop_binCRTT_muon2016M_bin4
71 prop_binCRF_electron2017_bin12_sm
72 prop_binSR_electron2018_bin13_sm
73 prop_binSR_muon2016M_bin13_sm
74 prop_binSR_electron2017_bin6_sm
75 prop_binSR_muon2016M_bin14_sm
76 prop_binCROS_muon2018_bin11
77 prop_binCRF_muon2016M_bin12_sm
78 prop_binCRF_muon2017_bin9_sm
79 prop_binCRTT_electron2018_bin1
80 prop_binCROS_muon2017_bin0
81 prop_binCRF_electron2016M_bin14_sm
82 prop_binCRF_electron2016M_bin15_sm
83 prop_binCROS_muon2018_bin13_WrongSign
84 prop_binCRF_muon2016M_bin8_sm
85 prop_binCRF_muon2016M_bin15_sm
86 prop_binCRF_muon2016M_bin11_sm
87 prop_binCROS_muon2018_bin10
88 prop_binCRTT_electron2018_bin0
89 jer_2017
90 prop_binCRF_muon2016M_bin11_WZ
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



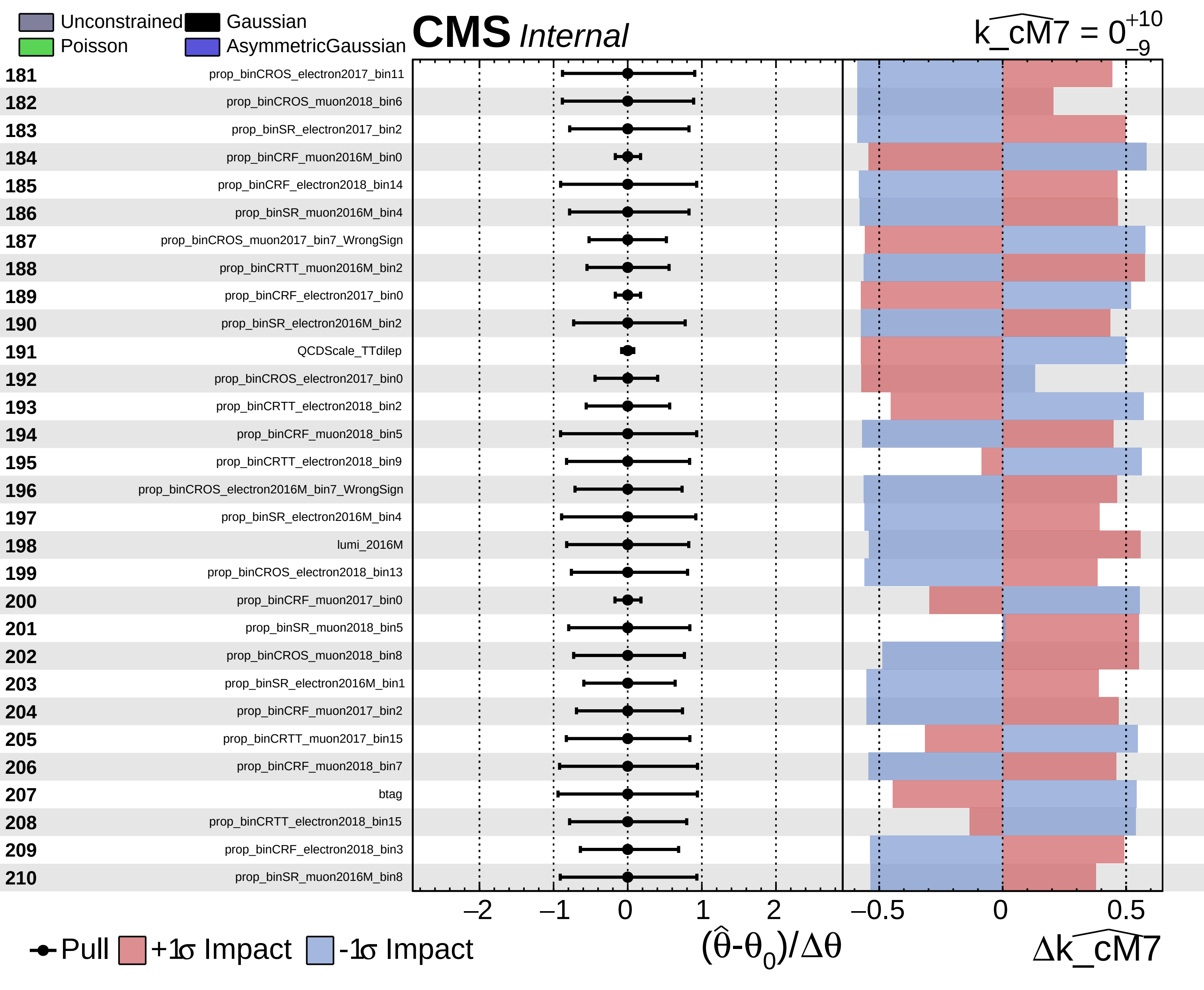
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
91 prop_binCRF_electron2016M_bin16
92 prop_binCROS_electron2017_bin8_WrongSign
93 prop_binCRF_electron2016M_bin15_TTTo2L2Nu
94 prop_binCROS_muon2016M_bin5_Fake
95 prop_binSR_electron2016M_bin0
96 prop_binCROS_muon2018_bin7_WrongSign
97 prop_binCRTT_muon2017_bin13
98 prop_binCRTT_electron2018_bin7
99 prop_binCRTT_electron2016M_bin1
100 prop_binCRTT_electron2016M_bin10
101 prop_binCRF_electron2017_bin14_sm
102 prop_binCRTT_muon2016M_bin3
103 prop_binCROS_muon2017_bin11_WrongSign
104 prop_binCRF_electron2016M_bin12
105 prop_binCRTT_electron2017_bin11
106 prop_binCRTT_muon2017_bin4
107 prop_binCRF_electron2016M_bin7
108 prop_binCRF_electron2016M_bin13
109 prop_binCROS_electron2016M_bin1
110 prop_binCRTT_muon2017_bin2
111 prop_binCRTT_electron2016M_bin4
112 prop_binCRF_electron2016M_bin8
113 prop_binCROS_muon2018_bin12_WrongSign
114 lumi_2017
115 prop_binCRF_electron2017_bin5
116 prop_binCRF_muon2016M_bin7
117 prop_binCRF_electron2017_bin10
118 prop_binCRF_electron2018_bin1
119 prop_binCROS_muon2018_bin16_WrongSign
120 prop_binCRTT_muon2017_bin10
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



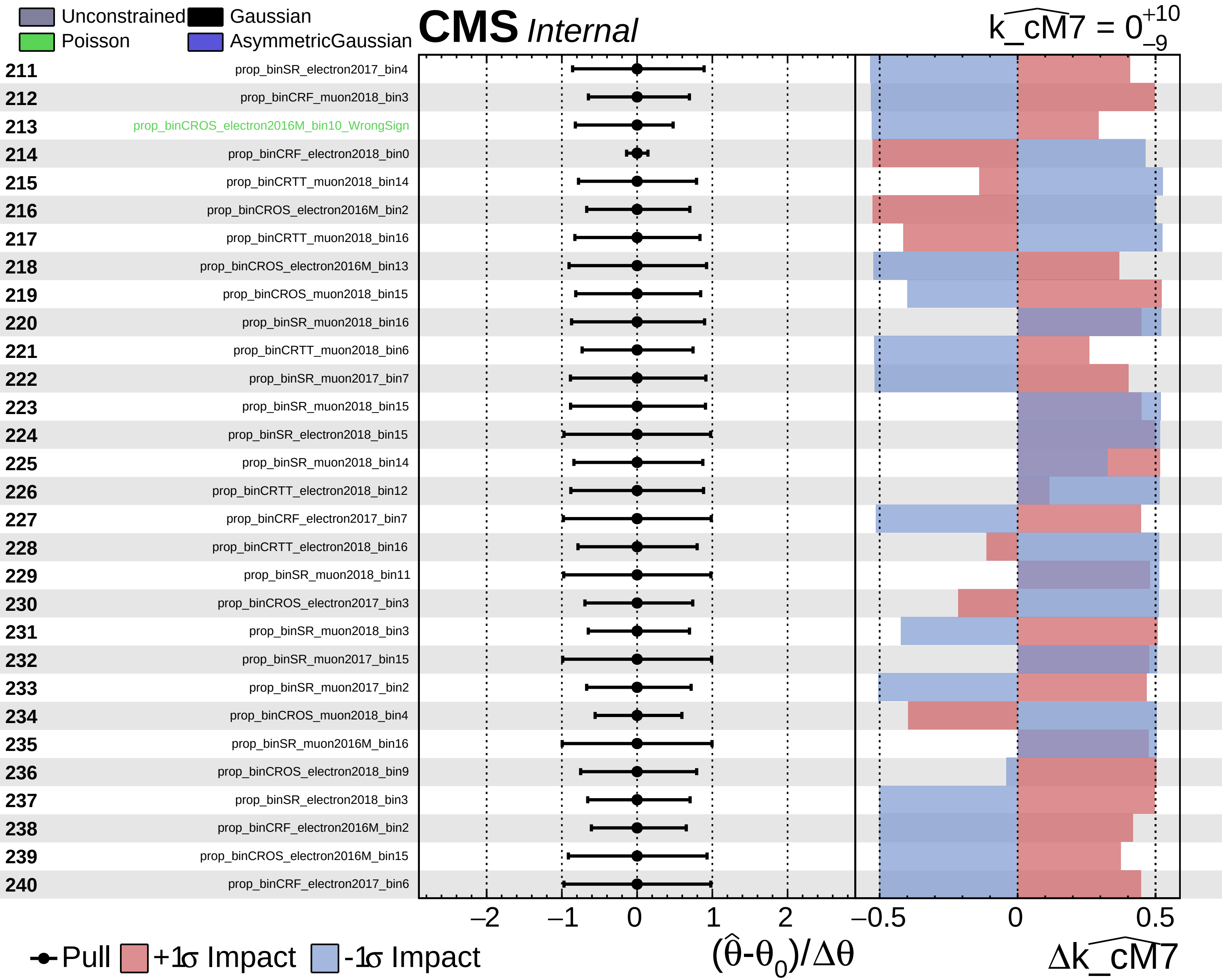
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0+10−9
121 prop_binCRF_electron2016M_bin15_Fake
122 prop_binCRF_muon2017_bin8
123 QCDScale_TVX
124 prop_binSR_electron2016M_bin6
125 prop_binCRF_muon2017_bin5
126 prop_binCRF_muon2017_bin13
127 prop_binCRF_muon2018_bin12
128 prop_binCRTT_muon2018_bin13
129 prop_binCRF_electron2017_bin4
130 prop_binCRTT_muon2018_bin2
131 metUnclust
132 prop_binCROS_electron2017_bin1
133 prop_binCRF_muon2016M_bin4
134 prop_binCROS_muon2016M_bin6
135 prop_binCRF_muon2017_bin4
136 prop_binCRF_muon2016M_bin6
137 prop_binSR_muon2016M_bin6
138 prop_binCROS_electron2017_bin10_WrongSign
139 prop_binCRF_electron2016M_bin10
140 prop_binCRF_electron2016M_bin9
141 prop_binCROS_muon2016M_bin7_Fake
142 prop_binCRTT_muon2018_bin4
143 prop_binSR_muon2018_bin10
144 prop_binCRF_electron2018_bin16
145 prop_binCROS_muon2016M_bin5_WrongSign
146 prop_binSR_muon2018_bin8
147 prop_binCROS_muon2018_bin3
148 prop_binCRF_muon2018_bin6
149 prop_binCRF_muon2016M_bin15_TTTo2L2Nu
150 prop_binCRF_muon2017_bin11
−2 −1 0 1 2
−0.5 0 0.5
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



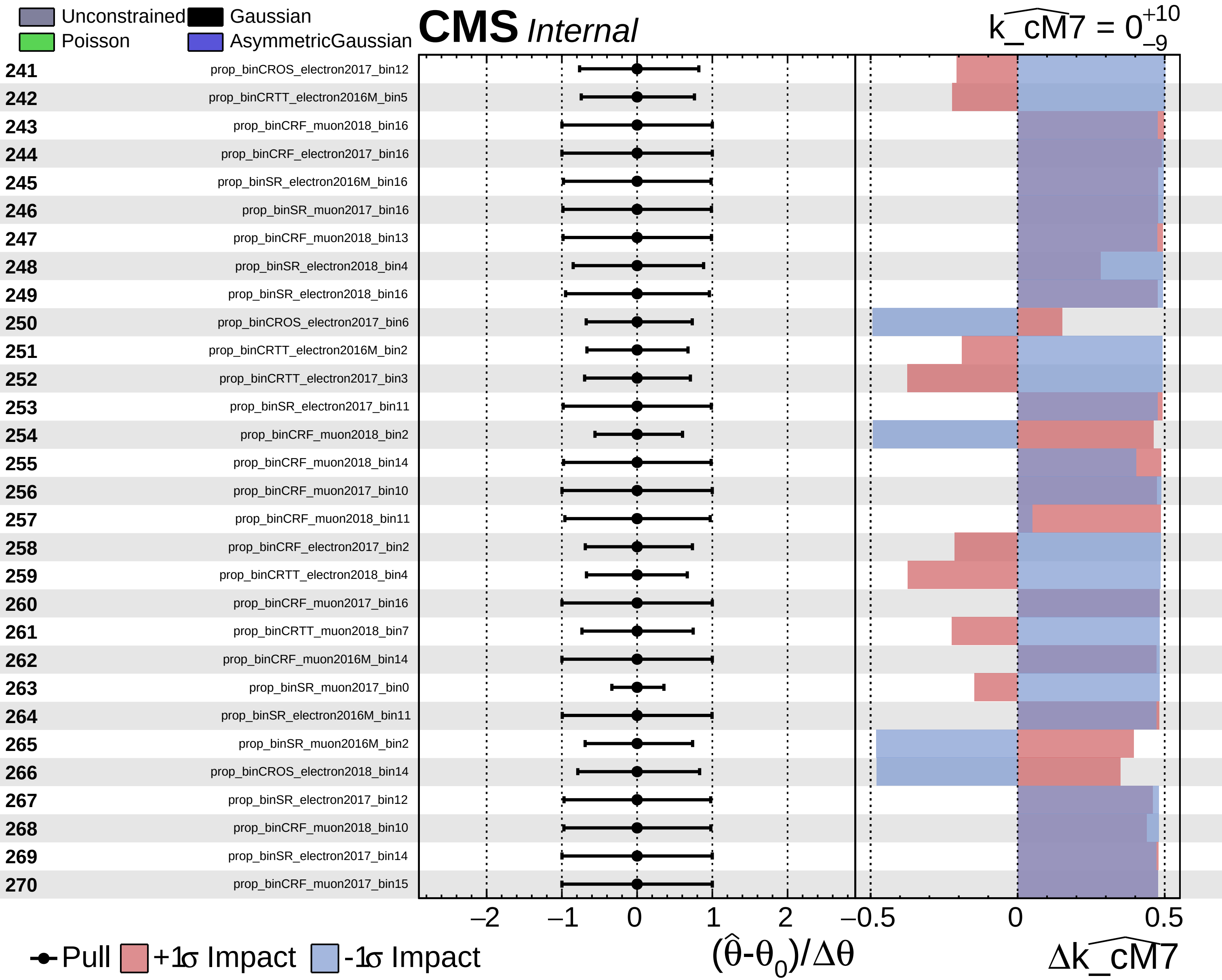
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
151 QCDScale_WpWpJJ_QCD
152 prop_binCROS_electron2016M_bin8
153 prop_binCRF_electron2018_bin11
154 prop_binSR_electron2016M_bin10
155 prop_binCRF_muon2016M_bin3
156 prop_binCRF_muon2018_bin9
157 prop_binCRF_electron2017_bin8
158 prop_binCRF_electron2017_bin13
159 prop_binSR_electron2017_bin5
160 prop_binSR_electron2016M_bin8
161 prop_binCRF_muon2016M_bin8_TTTo2L2Nu
162 prop_binSR_muon2016M_bin5
163 prop_binCROS_electron2017_bin9_WrongSign
164 prop_binCRF_electron2016M_bin0
165 prop_binCROS_electron2016M_bin12_WrongSign
166 prop_binCRF_electron2018_bin4
167 prop_binCRF_muon2018_bin4
168 prop_binCRTT_electron2017_bin8
169 prop_binSR_electron2016M_bin3
170 lumi_2018
171 prop_binCROS_muon2017_bin15_WrongSign
172 prop_binCROS_muon2016M_bin11
173 prop_binCROS_electron2018_bin6_WrongSign
174 prop_binCROS_muon2017_bin4
175 prop_binCRF_muon2018_bin0
176 mistag
177 prop_binCROS_muon2017_bin5_WrongSign
178 prop_binCRTT_muon2017_bin0
179 tau_vsjet_2016M
180 prop_binCRF_muon2016M_bin2
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



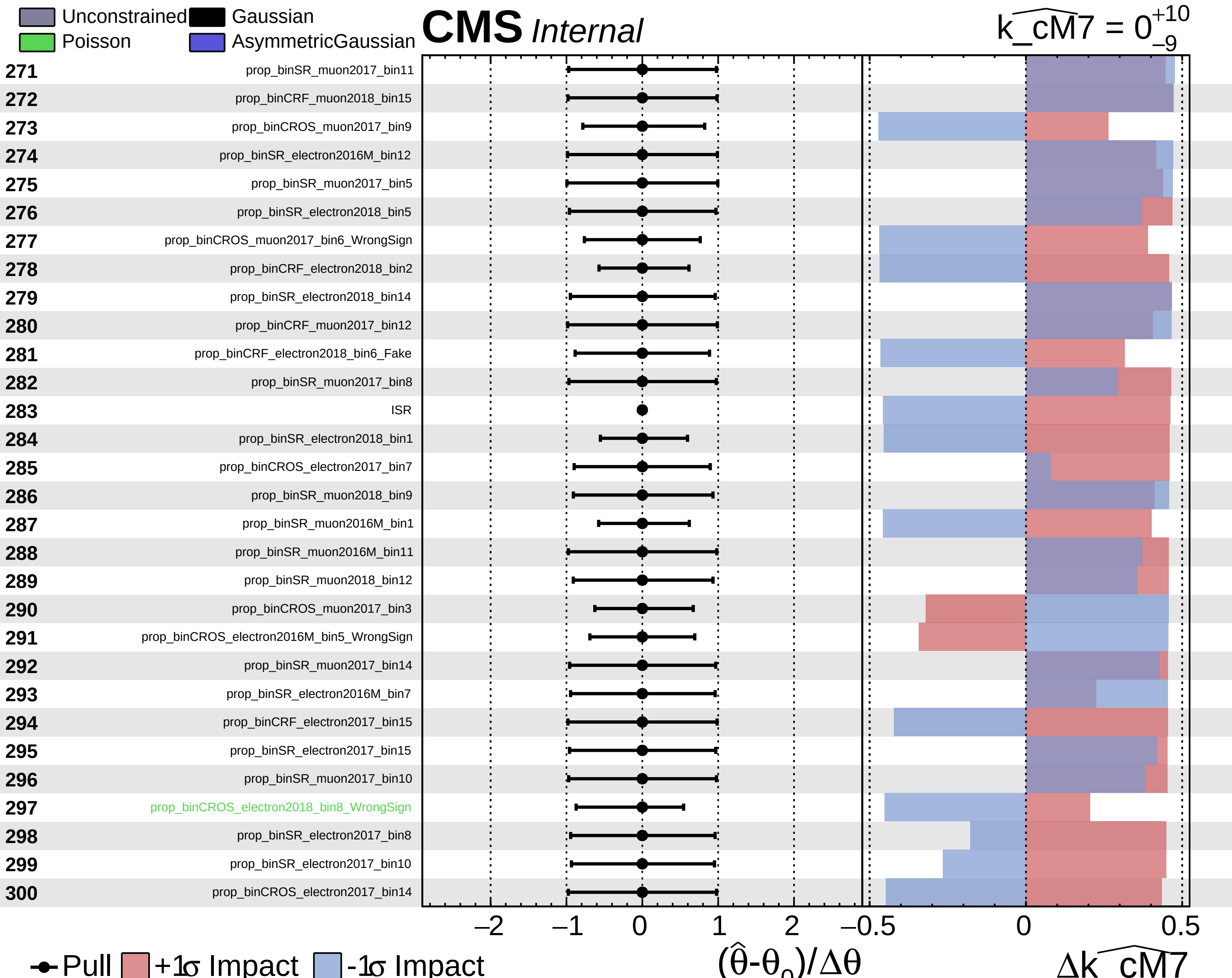
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +10 −9
181 prop_binCROS_electron2017_bin11
182 prop_binCROS_muon2018_bin6
183 prop_binSR_electron2017_bin2
184 prop_binCRF_muon2016M_bin0
185 prop_binCRF_electron2018_bin14
186 prop_binSR_muon2016M_bin4
187 prop_binCROS_muon2017_bin7_WrongSign
188 prop_binCRTT_muon2016M_bin2
189 prop_binCRF_electron2017_bin0
190 prop_binSR_electron2016M_bin2
191 QCDScale_TTdilep
192 prop_binCROS_electron2017_bin0
193 prop_binCRTT_electron2018_bin2
194 prop_binCRF_muon2018_bin5
195 prop_binCRTT_electron2018_bin9
196 prop_binCROS_electron2016M_bin7_WrongSign
197 prop_binSR_electron2016M_bin4
198 lumi_2016M
199 prop_binCROS_electron2018_bin13
200 prop_binCRF_muon2017_bin0
201 prop_binSR_muon2018_bin5
202 prop_binCROS_muon2018_bin8
203 prop_binSR_electron2016M_bin1
204 prop_binCRF_muon2017_bin2
205 prop_binCRTT_muon2017_bin15
206 prop_binCRF_muon2018_bin7
207 btag
208 prop_binCRTT_electron2018_bin15
209 prop_binCRF_electron2018_bin3
210 prop_binSR_muon2016M_bin8
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



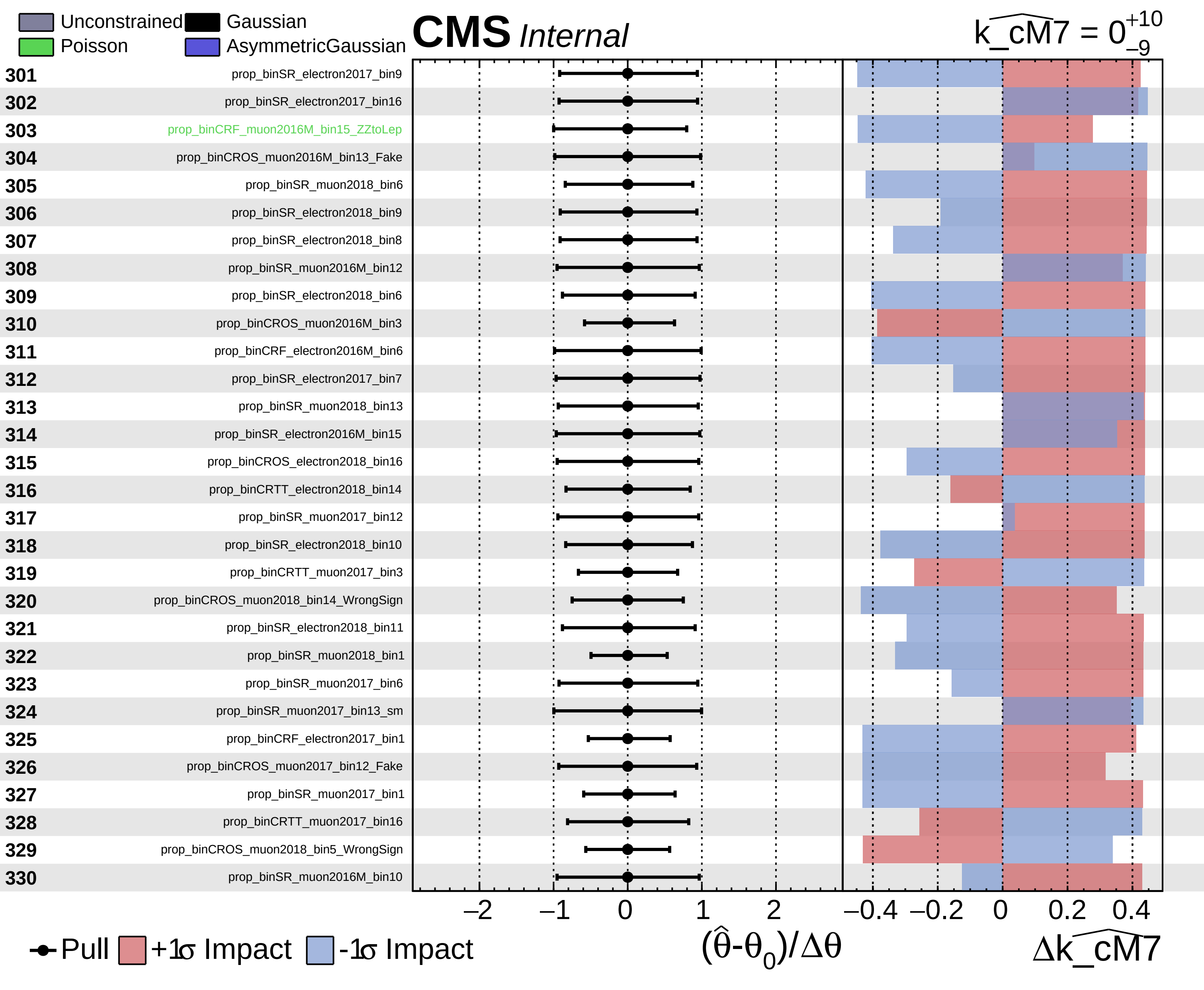
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
211 prop_binSR_electron2017_bin4
212 prop_binCRF_muon2018_bin3
213 prop_binCROS_electron2016M_bin10_WrongSign
214 prop_binCRF_electron2018_bin0
215 prop_binCRTT_muon2018_bin14
216 prop_binCROS_electron2016M_bin2
217 prop_binCRTT_muon2018_bin16
218 prop_binCROS_electron2016M_bin13
219 prop_binCROS_muon2018_bin15
220 prop_binSR_muon2018_bin16
221 prop_binCRTT_muon2018_bin6
222 prop_binSR_muon2017_bin7
223 prop_binSR_muon2018_bin15
224 prop_binSR_electron2018_bin15
225 prop_binSR_muon2018_bin14
226 prop_binCRTT_electron2018_bin12
227 prop_binCRF_electron2017_bin7
228 prop_binCRTT_electron2018_bin16
229 prop_binSR_muon2018_bin11
230 prop_binCROS_electron2017_bin3
231 prop_binSR_muon2018_bin3
232 prop_binSR_muon2017_bin15
233 prop_binSR_muon2017_bin2
234 prop_binCROS_muon2018_bin4
235 prop_binSR_muon2016M_bin16
236 prop_binCROS_electron2018_bin9
237 prop_binSR_electron2018_bin3
238 prop_binCRF_electron2016M_bin2
239 prop_binCROS_electron2016M_bin15
240 prop_binCRF_electron2017_bin6
−2
−1
0
1
2
−0.5
0
0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



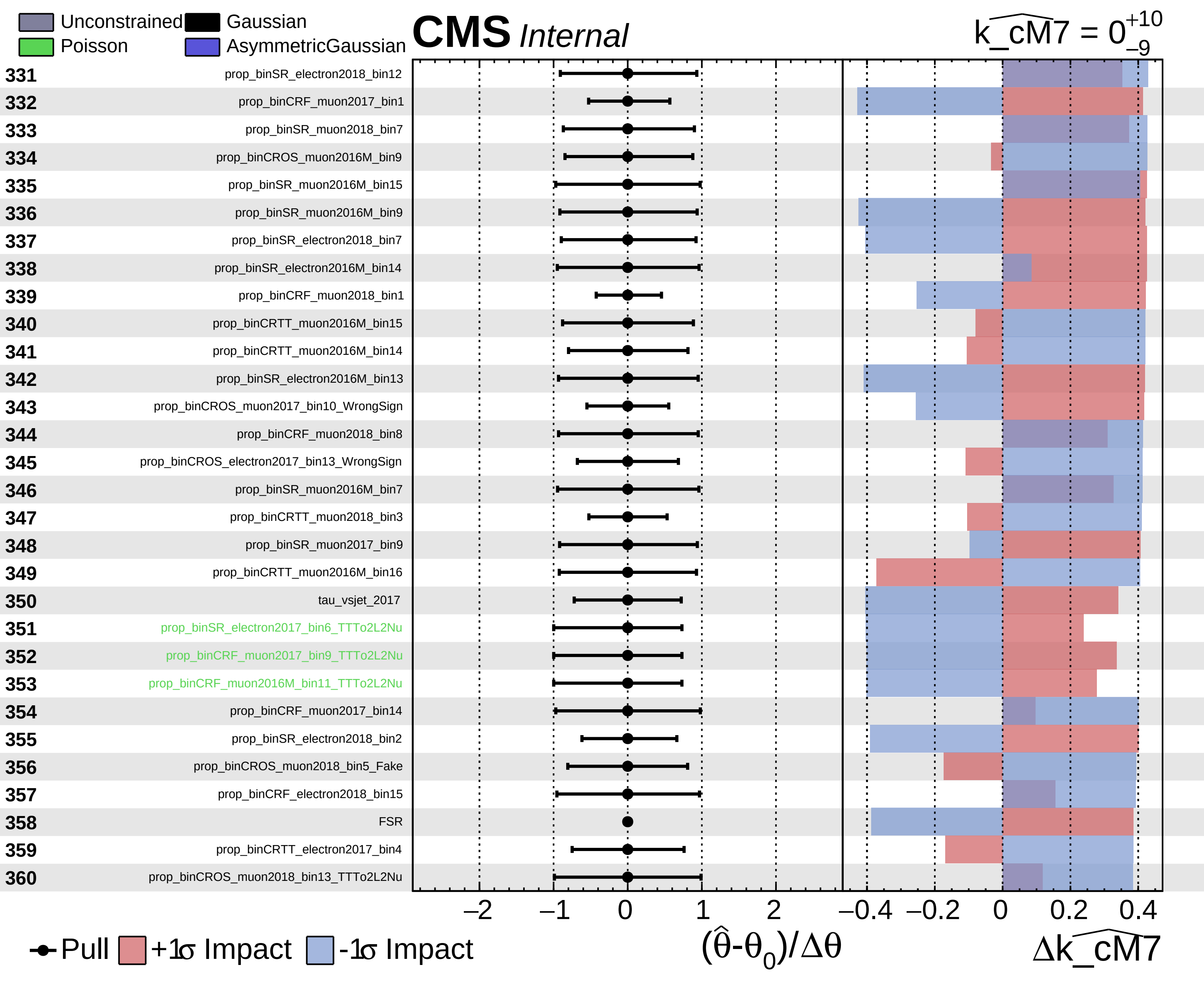
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
241 prop_binCROS_electron2017_bin12
242 prop_binCRTT_electron2016M_bin5
243 prop_binCRF_muon2018_bin16
244 prop_binCRF_electron2017_bin16
245 prop_binSR_electron2016M_bin16
246 prop_binSR_muon2017_bin16
247 prop_binCRF_muon2018_bin13
248 prop_binSR_electron2018_bin4
249 prop_binSR_electron2018_bin16
250 prop_binCROS_electron2017_bin6
251 prop_binCRTT_electron2016M_bin2
252 prop_binCRTT_electron2017_bin3
253 prop_binSR_electron2017_bin11
254 prop_binCRF_muon2018_bin2
255 prop_binCRF_muon2018_bin14
256 prop_binCRF_muon2017_bin10
257 prop_binCRF_muon2018_bin11
258 prop_binCRF_electron2017_bin2
259 prop_binCRTT_electron2018_bin4
260 prop_binCRF_muon2017_bin16
261 prop_binCRTT_muon2018_bin7
262 prop_binCRF_muon2016M_bin14
263 prop_binSR_muon2017_bin0
264 prop_binSR_electron2016M_bin11
265 prop_binSR_muon2016M_bin2
266 prop_binCROS_electron2018_bin14
267 prop_binSR_electron2017_bin12
268 prop_binCRF_muon2018_bin10
269 prop_binSR_electron2017_bin14
270 prop_binCRF_muon2017_bin15
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



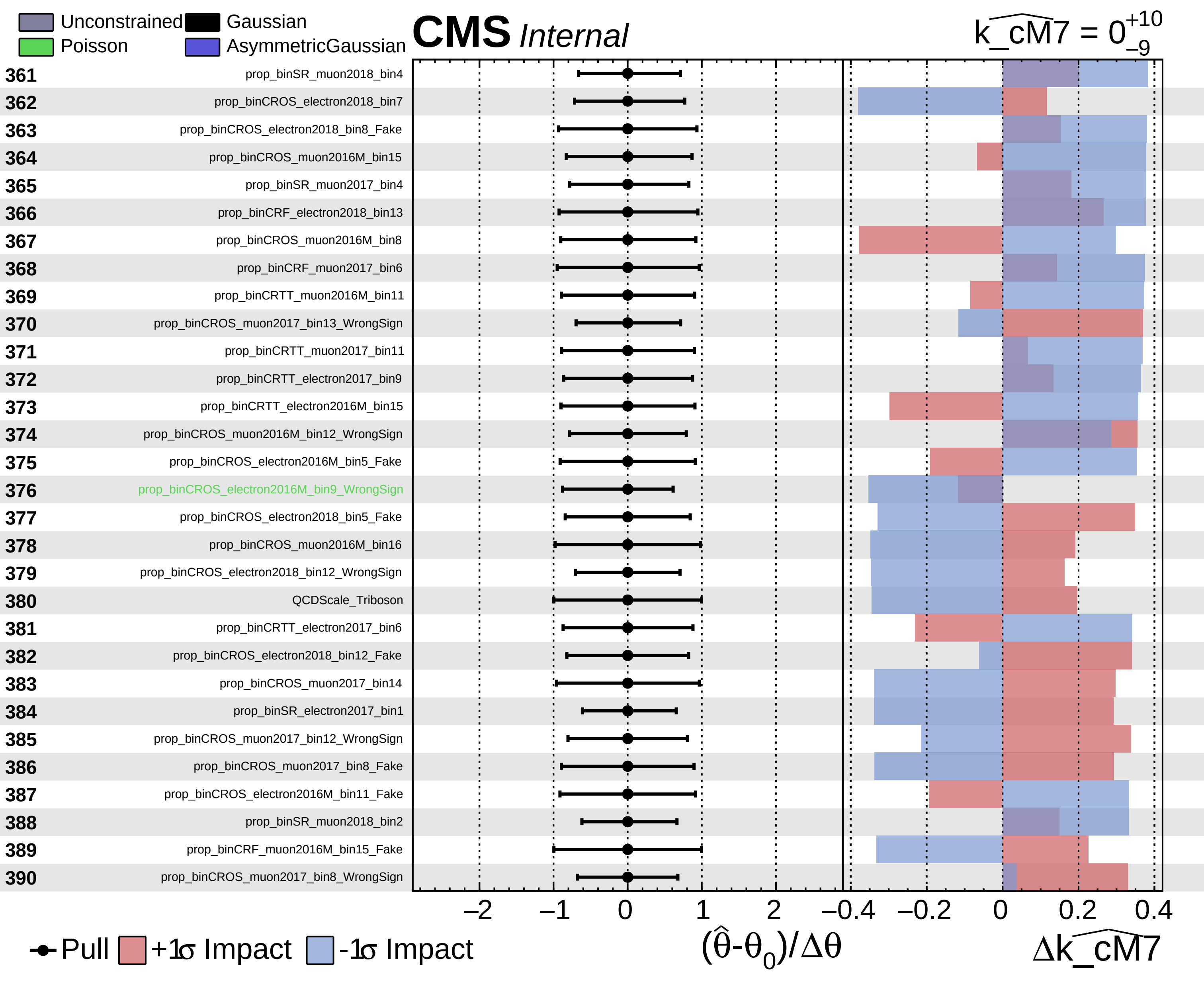
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0^{+10}_{-9}
271 prop_binSR_muon2017_bin11
272 prop_binCRF_muon2018_bin15
273 prop_binCROS_muon2017_bin9
274 prop_binSR_electron2016M_bin12
275 prop_binSR_muon2017_bin5
276 prop_binSR_electron2018_bin5
277 prop_binCROS_muon2017_bin6_WrongSign
278 prop_binCRF_electron2018_bin2
279 prop_binSR_electron2018_bin14
280 prop_binCRF_muon2017_bin12
281 prop_binCRF_electron2018_bin6_Fake
282 prop_binSR_muon2017_bin8
283 ISR
284 prop_binSR_electron2018_bin1
285 prop_binCROS_electron2017_bin7
286 prop_binSR_muon2018_bin9
287 prop_binSR_muon2016M_bin1
288 prop_binSR_muon2016M_bin11
289 prop_binSR_muon2018_bin12
290 prop_binCROS_muon2017_bin3
291 prop_binCROS_electron2016M_bin5_WrongSign
292 prop_binSR_muon2017_bin14
293 prop_binSR_electron2016M_bin7
294 prop_binCRF_electron2017_bin15
295 prop_binSR_electron2017_bin15
296 prop_binSR_muon2017_bin10
297 prop_binCROS_electron2018_bin8_WrongSign
298 prop_binSR_electron2017_bin8
299 prop_binSR_electron2017_bin10
300 prop_binCROS_electron2017_bin14
−2 −1 0 1 2
−0.5 0 0.5
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



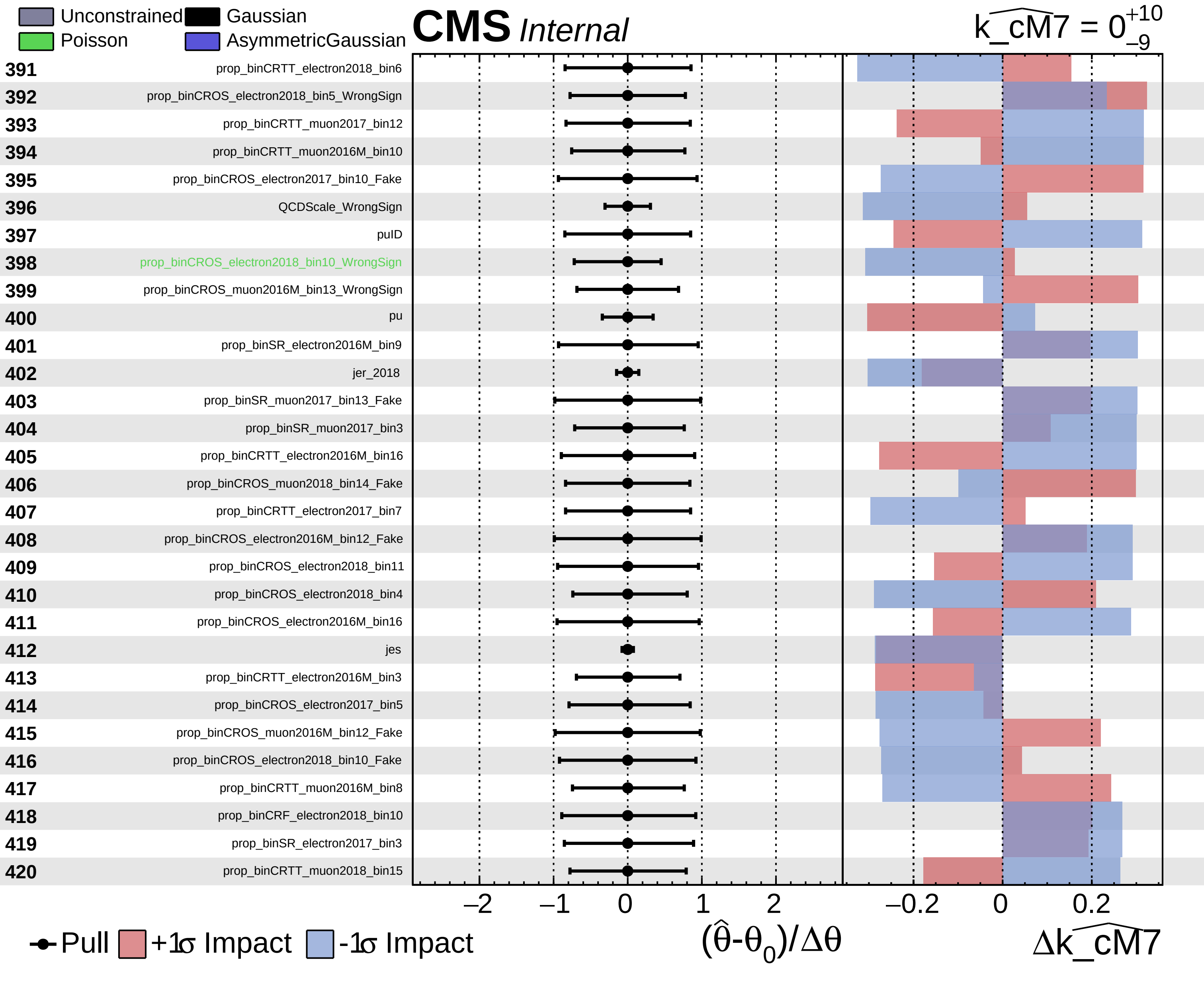
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +10 −9
301 prop_binSR_electron2017_bin9
302 prop_binSR_electron2017_bin16
303 prop_binCRF_muon2016M_bin15_ZZtoLep
304 prop_binCROS_muon2016M_bin13_Fake
305 prop_binSR_muon2018_bin6
306 prop_binSR_electron2018_bin9
307 prop_binSR_electron2018_bin8
308 prop_binSR_muon2016M_bin12
309 prop_binSR_electron2018_bin6
310 prop_binCROS_muon2016M_bin3
311 prop_binCRF_electron2016M_bin6
312 prop_binSR_electron2017_bin7
313 prop_binSR_muon2018_bin13
314 prop_binSR_electron2016M_bin15
315 prop_binCROS_electron2018_bin16
316 prop_binCRTT_electron2018_bin14
317 prop_binSR_muon2017_bin12
318 prop_binSR_electron2018_bin10
319 prop_binCRTT_muon2017_bin3
320 prop_binCROS_muon2018_bin14_WrongSign
321 prop_binSR_electron2018_bin11
322 prop_binSR_muon2018_bin1
323 prop_binSR_muon2017_bin6
324 prop_binSR_muon2017_bin13_sm
325 prop_binCRF_electron2017_bin1
326 prop_binCROS_muon2017_bin12_Fake
327 prop_binSR_muon2017_bin1
328 prop_binCRTT_muon2017_bin16
329 prop_binCROS_muon2018_bin5_WrongSign
330 prop_binSR_muon2016M_bin10
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7



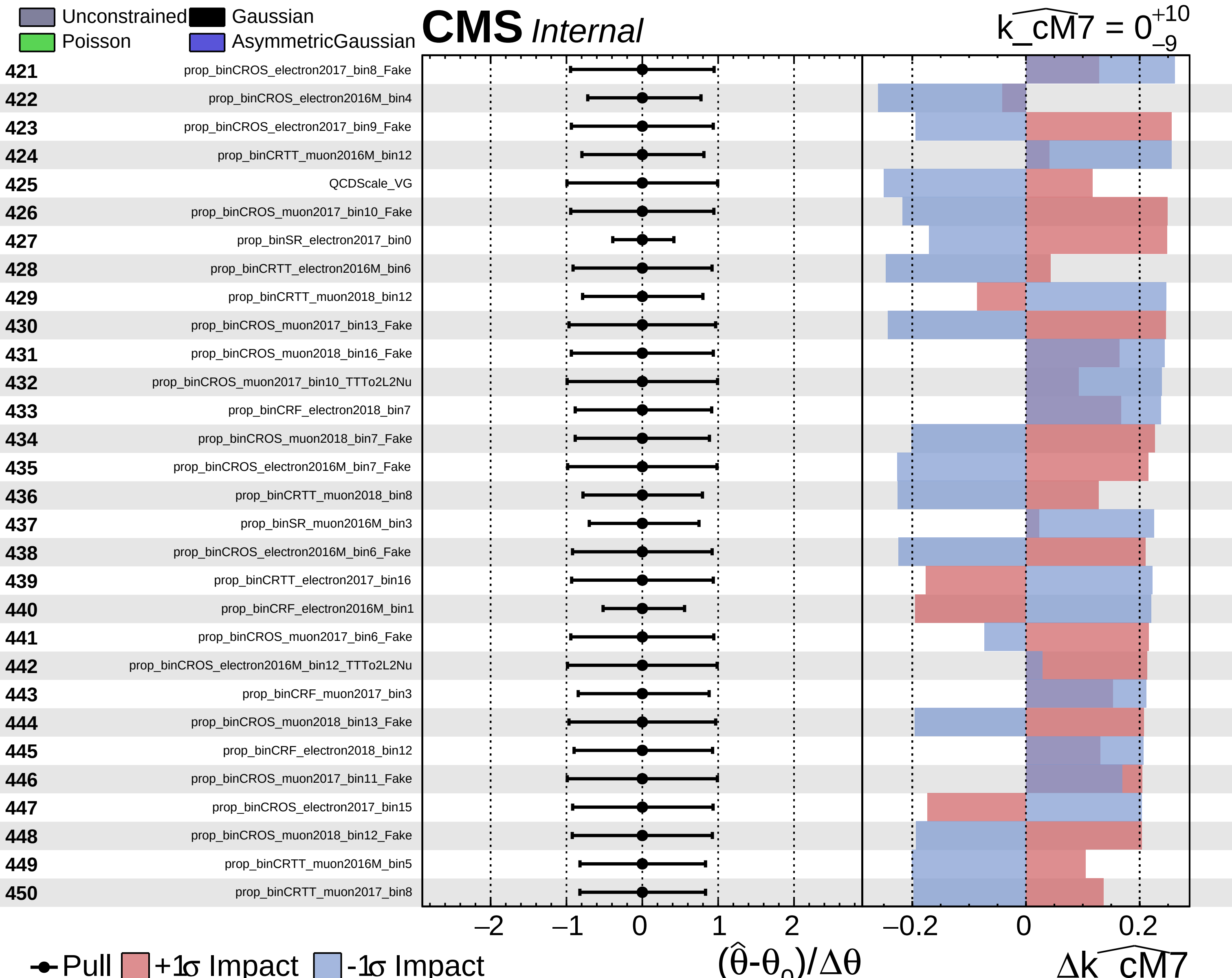
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0+10 −9
331 prop_binSR_electron2018_bin12
332 prop_binCRF_muon2017_bin1
333 prop_binSR_muon2018_bin7
334 prop_binCROS_muon2016M_bin9
335 prop_binSR_muon2016M_bin15
336 prop_binSR_muon2016M_bin9
337 prop_binSR_electron2018_bin7
338 prop_binSR_electron2016M_bin14
339 prop_binCRF_muon2018_bin1
340 prop_binCRTT_muon2016M_bin15
341 prop_binCRTT_muon2016M_bin14
342 prop_binSR_electron2016M_bin13
343 prop_binCROS_muon2017_bin10_WrongSign
344 prop_binCRF_muon2018_bin8
345 prop_binCROS_electron2017_bin13_WrongSign
346 prop_binSR_muon2016M_bin7
347 prop_binCRTT_muon2018_bin3
348 prop_binSR_muon2017_bin9
349 prop_binCRTT_muon2016M_bin16
350 tau_vsjet_2017
351 prop_binSR_electron2017_bin6_TTTo2L2Nu
352 prop_binCRF_muon2017_bin9_TTTo2L2Nu
353 prop_binCRF_muon2016M_bin11_TTTo2L2Nu
354 prop_binCRF_muon2017_bin14
355 prop_binSR_electron2018_bin2
356 prop_binCROS_muon2018_bin5_Fake
357 prop_binCRF_electron2018_bin15
358 FSR
359 prop_binCRTT_electron2017_bin4
360 prop_binCROS_muon2018_bin13_TTTo2L2Nu
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7

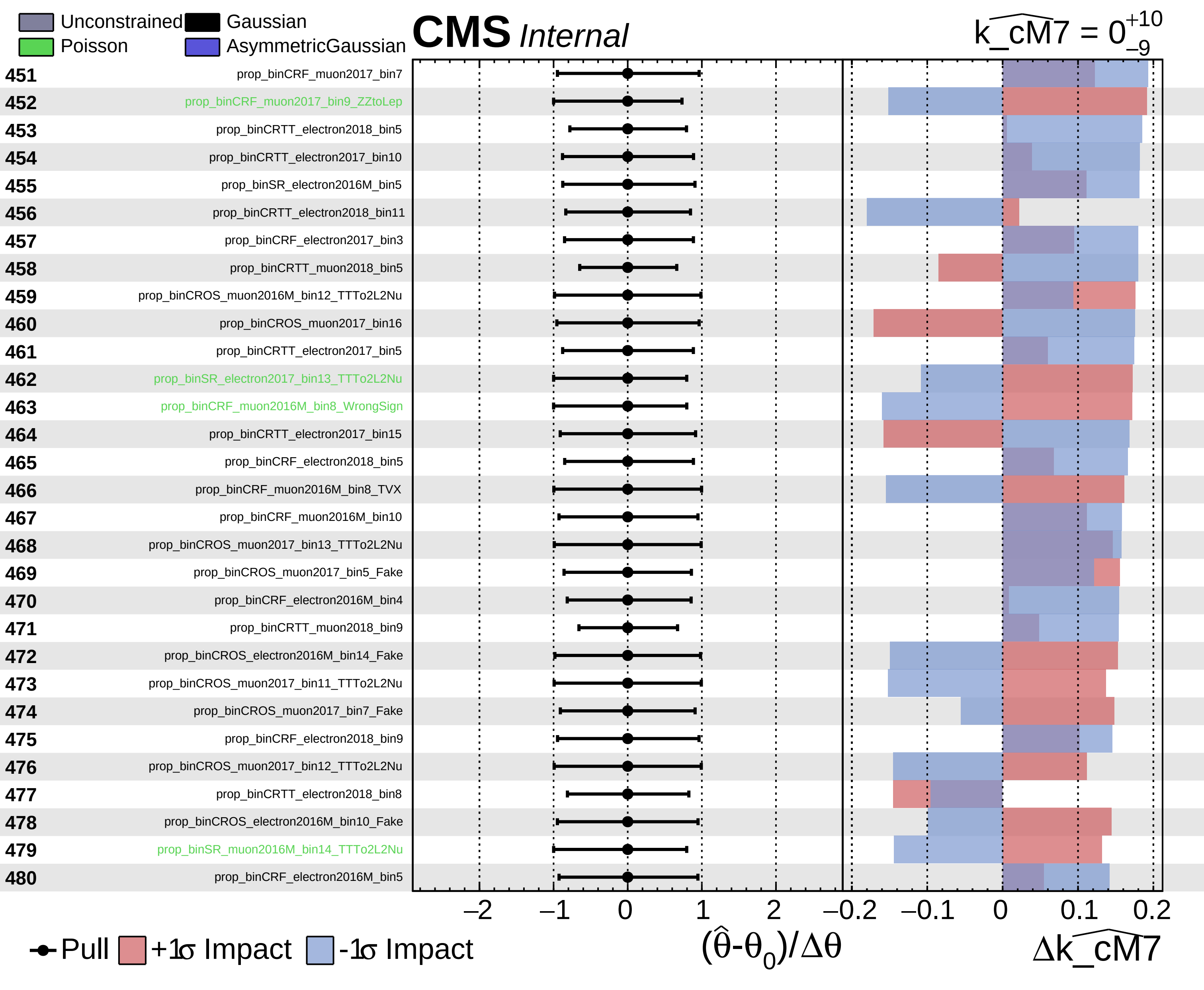


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0^{+10}_{-9}
361 prop_binSR_muon2018_bin4
362 prop_binCROS_electron2018_bin7
363 prop_binCROS_electron2018_bin8_Fake
364 prop_binCROS_muon2016M_bin15
365 prop_binSR_muon2017_bin4
366 prop_binCRF_electron2018_bin13
367 prop_binCROS_muon2016M_bin8
368 prop_binCRF_muon2017_bin6
369 prop_binCRTT_muon2016M_bin11
370 prop_binCROS_muon2017_bin13_WrongSign
371 prop_binCRTT_muon2017_bin11
372 prop_binCRTT_electron2017_bin9
373 prop_binCRTT_electron2016M_bin15
374 prop_binCROS_muon2016M_bin12_WrongSign
375 prop_binCROS_electron2016M_bin5_Fake
376 prop_binCROS_electron2016M_bin9_WrongSign
377 prop_binCROS_electron2018_bin5_Fake
378 prop_binCROS_muon2016M_bin16
379 prop_binCROS_electron2018_bin12_WrongSign
380 QCDScale_Triboson
381 prop_binCRTT_electron2017_bin6
382 prop_binCROS_electron2018_bin12_Fake
383 prop_binCROS_muon2017_bin14
384 prop_binSR_electron2017_bin1
385 prop_binCROS_muon2017_bin12_WrongSign
386 prop_binCROS_muon2017_bin8_Fake
387 prop_binCROS_electron2016M_bin11_Fake
388 prop_binSR_muon2018_bin2
389 prop_binCRF_muon2016M_bin15_Fake
390 prop_binCROS_muon2017_bin8_WrongSign
−2
−1
0
1
2
−0.4
−0.2
0
0.2
0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$

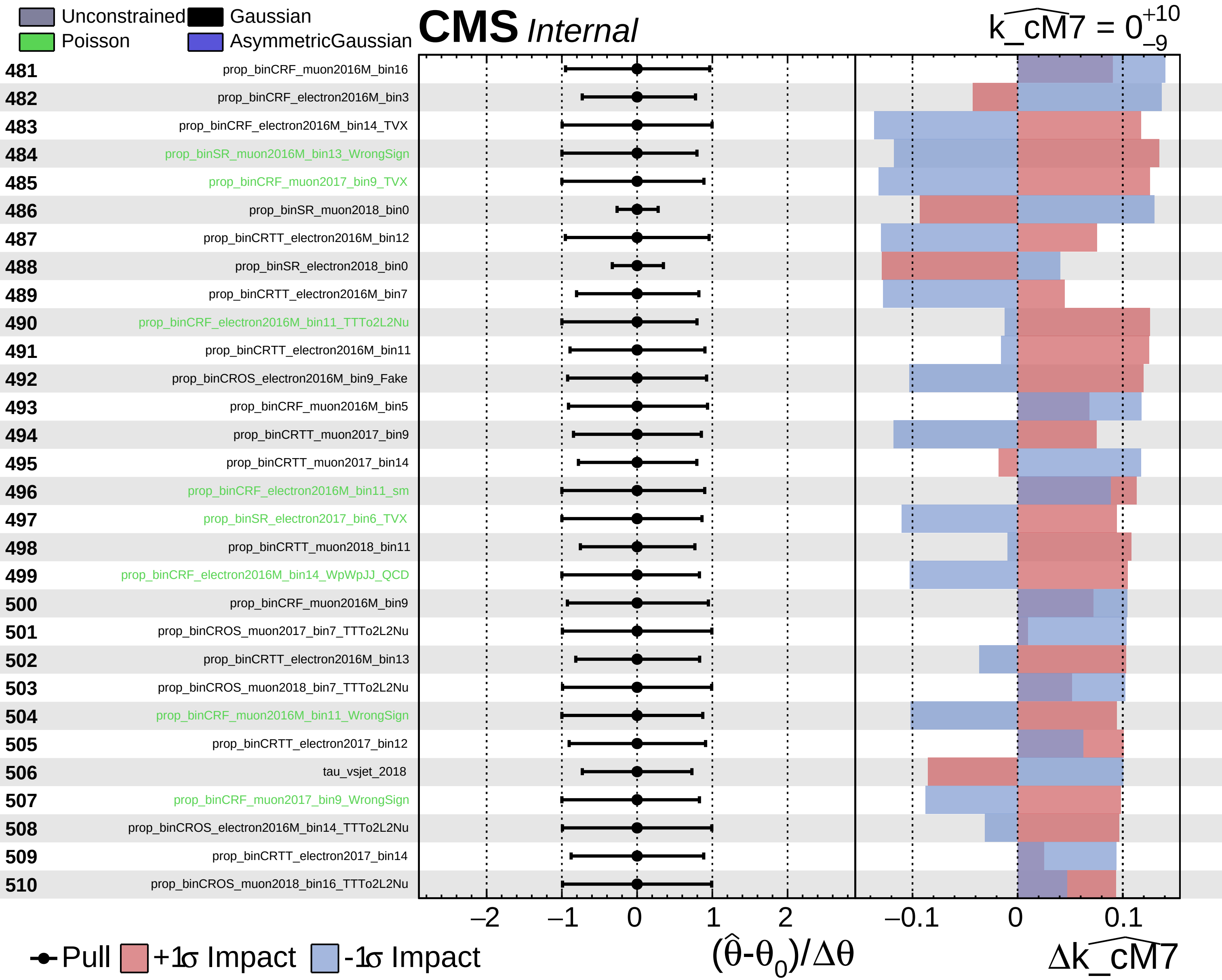


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
391 prop_binCRTT_electron2018_bin6
392 prop_binCROS_electron2018_bin5_WrongSign
393 prop_binCRTT_muon2017_bin12
394 prop_binCRTT_muon2016M_bin10
395 prop_binCROS_electron2017_bin10_Fake
396 QCDScale_WrongSign
397 puID
398 prop_binCROS_electron2018_bin10_WrongSign
399 prop_binCROS_muon2016M_bin13_WrongSign
400 pu
401 prop_binSR_electron2016M_bin9
402 jer_2018
403 prop_binSR_muon2017_bin13_Fake
404 prop_binSR_muon2017_bin3
405 prop_binCRTT_electron2016M_bin16
406 prop_binCROS_muon2018_bin14_Fake
407 prop_binCRTT_electron2017_bin7
408 prop_binCROS_electron2016M_bin12_Fake
409 prop_binCROS_electron2018_bin11
410 prop_binCROS_electron2018_bin4
411 prop_binCROS_electron2016M_bin16
412 jes
413 prop_binCRTT_electron2016M_bin3
414 prop_binCROS_electron2017_bin5
415 prop_binCROS_muon2016M_bin12_Fake
416 prop_binCROS_electron2018_bin10_Fake
417 prop_binCRTT_muon2016M_bin8
418 prop_binCRF_electron2018_bin10
419 prop_binSR_electron2017_bin3
420 prop_binCRTT_muon2018_bin15
−2 −1 0 1 2
−0.2 0 0.2
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$

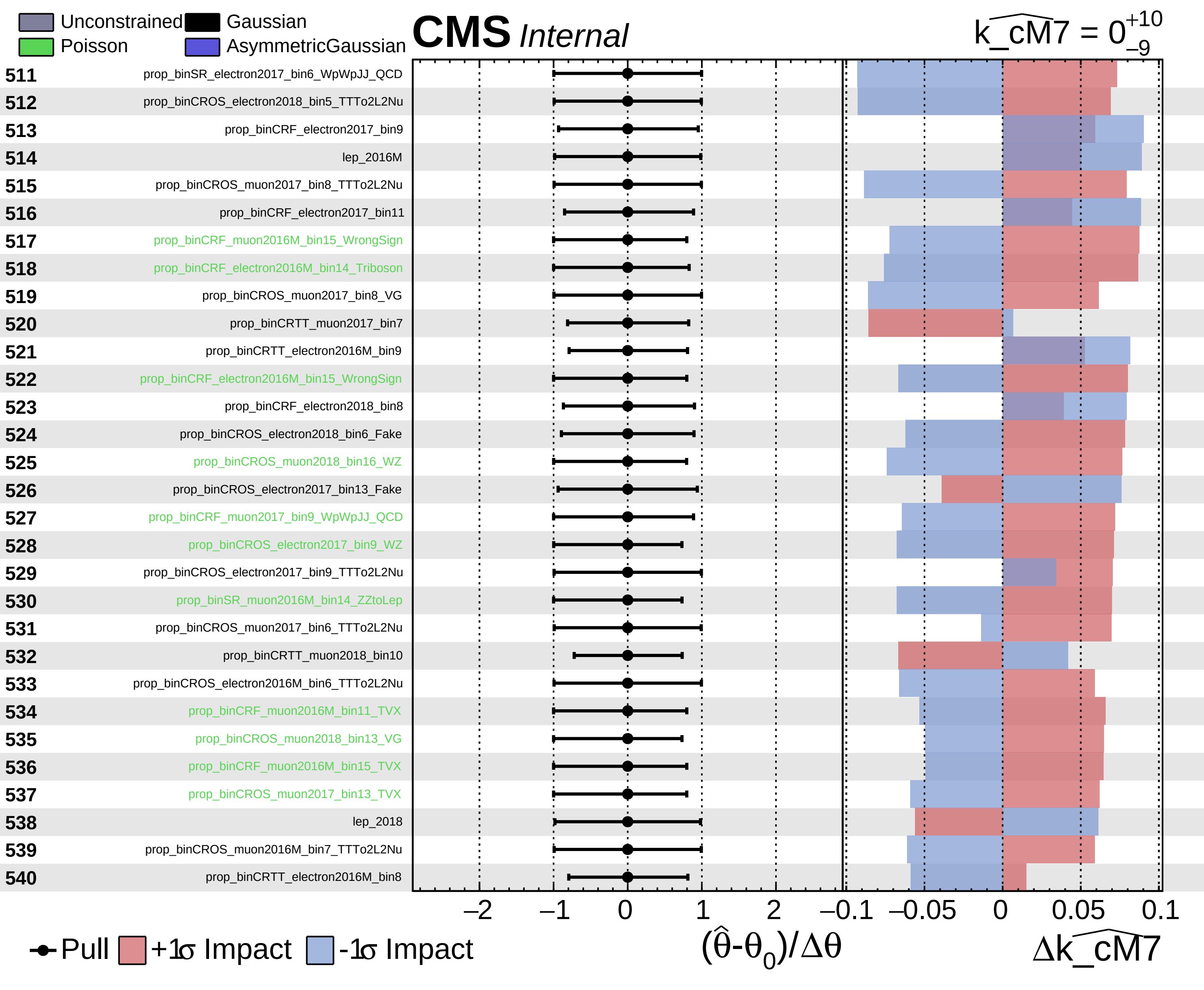

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0+10−9
421 prop_binCROS_electron2017_bin8_Fake
422 prop_binCROS_electron2016M_bin4
423 prop_binCROS_electron2017_bin9_Fake
424 prop_binCRTT_muon2016M_bin12
425 QCDScale_VG
426 prop_binCROS_muon2017_bin10_Fake
427 prop_binSR_electron2017_bin0
428 prop_binCRTT_electron2016M_bin6
429 prop_binCRTT_muon2018_bin12
430 prop_binCROS_muon2017_bin13_Fake
431 prop_binCROS_muon2018_bin16_Fake
432 prop_binCROS_muon2017_bin10_TTTo2L2Nu
433 prop_binCRF_electron2018_bin7
434 prop_binCROS_muon2018_bin7_Fake
435 prop_binCROS_electron2016M_bin7_Fake
436 prop_binCRTT_muon2018_bin8
437 prop_binSR_muon2016M_bin3
438 prop_binCROS_electron2016M_bin6_Fake
439 prop_binCRTT_electron2017_bin16
440 prop_binCRF_electron2016M_bin1
441 prop_binCROS_muon2017_bin6_Fake
442 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
443 prop_binCRF_muon2017_bin3
444 prop_binCROS_muon2018_bin13_Fake
445 prop_binCRF_electron2018_bin12
446 prop_binCROS_muon2017_bin11_Fake
447 prop_binCROS_electron2017_bin15
448 prop_binCROS_muon2018_bin12_Fake
449 prop_binCRTT_muon2016M_bin5
450 prop_binCRTT_muon2017_bin8
−2
−1
0
1
2
−0.2
0
0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



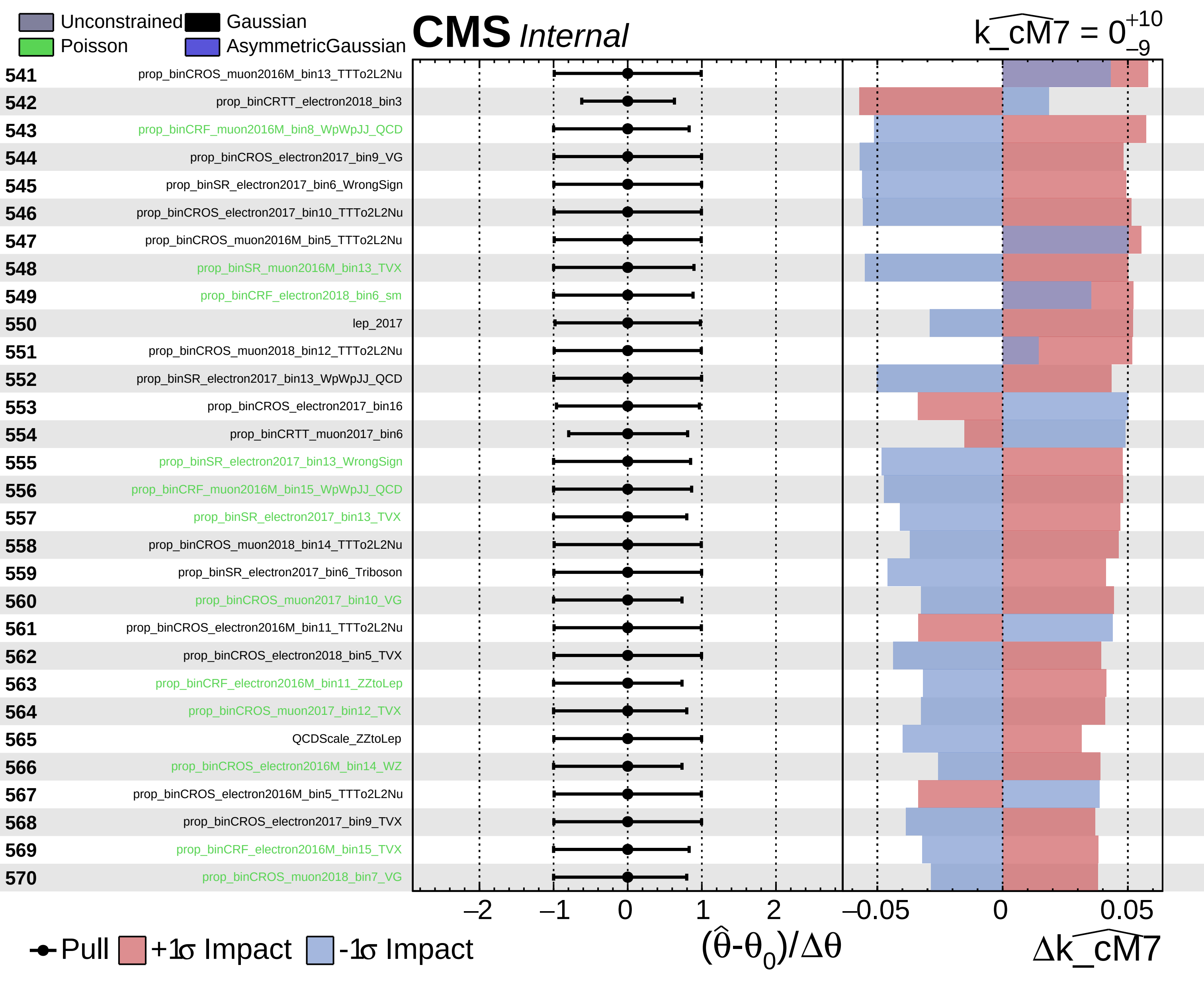
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
451 prop_binCRF_muon2017_bin7
452 prop_binCRF_muon2017_bin9_ZZtoLep
453 prop_binCRTT_electron2018_bin5
454 prop_binCRTT_electron2017_bin10
455 prop_binSR_electron2016M_bin5
456 prop_binCRTT_electron2018_bin11
457 prop_binCRF_electron2017_bin3
458 prop_binCRTT_muon2018_bin5
459 prop_binCROS_muon2016M_bin12_TTTo2L2Nu
460 prop_binCROS_muon2017_bin16
461 prop_binCRTT_electron2017_bin5
462 prop_binSR_electron2017_bin13_TTTo2L2Nu
463 prop_binCRF_muon2016M_bin8_WrongSign
464 prop_binCRTT_electron2017_bin15
465 prop_binCRF_electron2018_bin5
466 prop_binCRF_muon2016M_bin8_TVX
467 prop_binCRF_muon2016M_bin10
468 prop_binCROS_muon2017_bin13_TTTo2L2Nu
469 prop_binCROS_muon2017_bin5_Fake
470 prop_binCRF_electron2016M_bin4
471 prop_binCRTT_muon2018_bin9
472 prop_binCROS_electron2016M_bin14_Fake
473 prop_binCROS_muon2017_bin11_TTTo2L2Nu
474 prop_binCROS_muon2017_bin7_Fake
475 prop_binCRF_electron2018_bin9
476 prop_binCROS_muon2017_bin12_TTTo2L2Nu
477 prop_binCRTT_electron2018_bin8
478 prop_binCROS_electron2016M_bin10_Fake
479 prop_binSR_muon2016M_bin14_TTTo2L2Nu
480 prop_binCRF_electron2016M_bin5
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



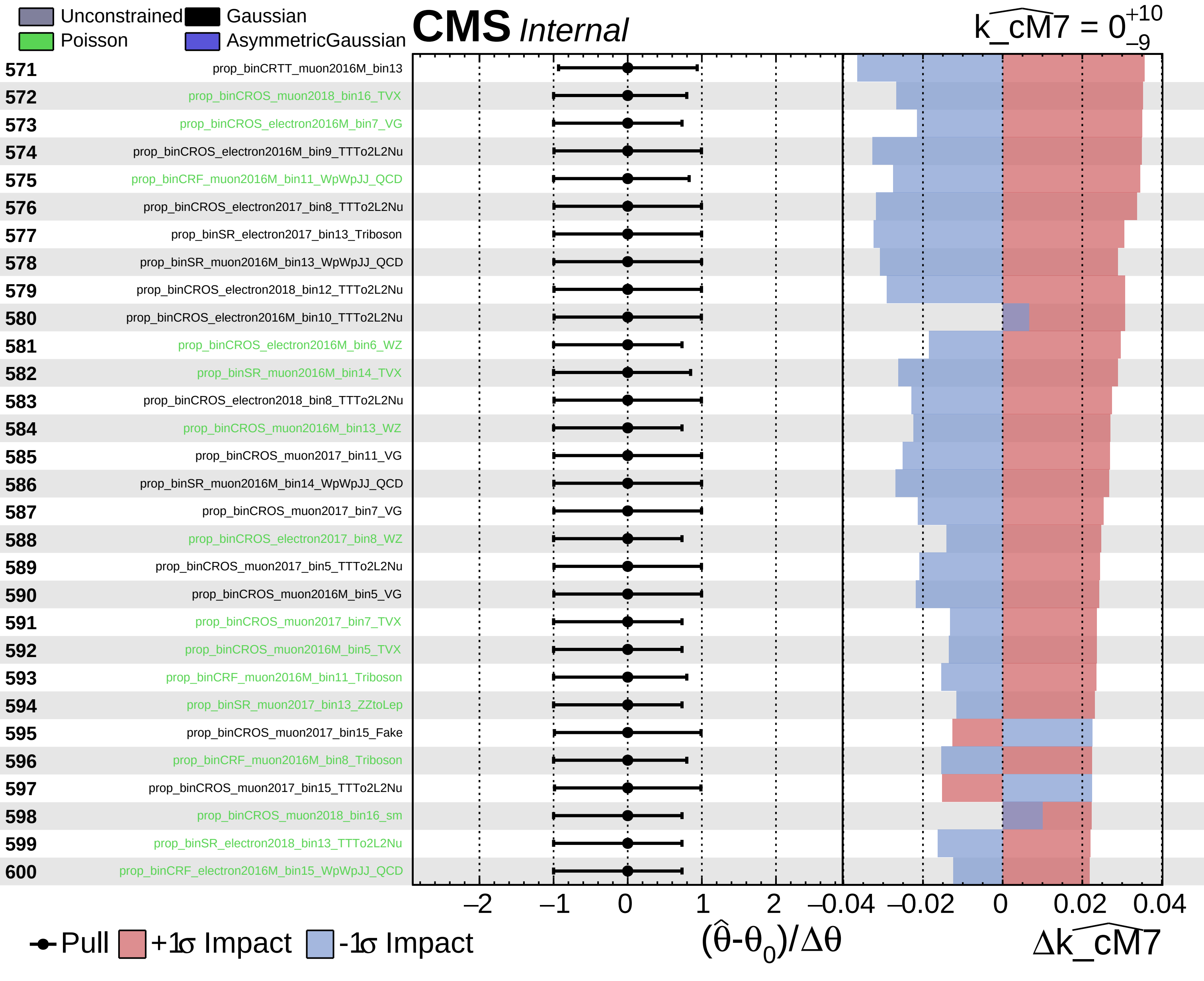
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
481 prop_binCRF_muon2016M_bin16
482 prop_binCRF_electron2016M_bin3
483 prop_binCRF_electron2016M_bin14_TVX
484 prop_binSR_muon2016M_bin13_WrongSign
485 prop_binCRF_muon2017_bin9_TVX
486 prop_binSR_muon2018_bin0
487 prop_binCRTT_electron2016M_bin12
488 prop_binSR_electron2018_bin0
489 prop_binCRTT_electron2016M_bin7
490 prop_binCRF_electron2016M_bin11_TTTo2L2Nu
491 prop_binCRTT_electron2016M_bin11
492 prop_binCROS_electron2016M_bin9_Fake
493 prop_binCRF_muon2016M_bin5
494 prop_binCRTT_muon2017_bin9
495 prop_binCRTT_muon2017_bin14
496 prop_binCRF_electron2016M_bin11_sm
497 prop_binSR_electron2017_bin6_TVX
498 prop_binCRTT_muon2018_bin11
499 prop_binCRF_electron2016M_bin14_WpWpJJ_QCD
500 prop_binCRF_muon2016M_bin9
501 prop_binCROS_muon2017_bin7_TTTo2L2Nu
502 prop_binCRTT_electron2016M_bin13
503 prop_binCROS_muon2018_bin7_TTTo2L2Nu
504 prop_binCRF_muon2016M_bin11_WrongSign
505 prop_binCRTT_electron2017_bin12
506 tau_vsjet_2018
507 prop_binCRF_muon2017_bin9_WrongSign
508 prop_binCROS_electron2016M_bin14_TTTo2L2Nu
509 prop_binCRTT_electron2017_bin14
510 prop_binCROS_muon2018_bin16_TTTo2L2Nu
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$

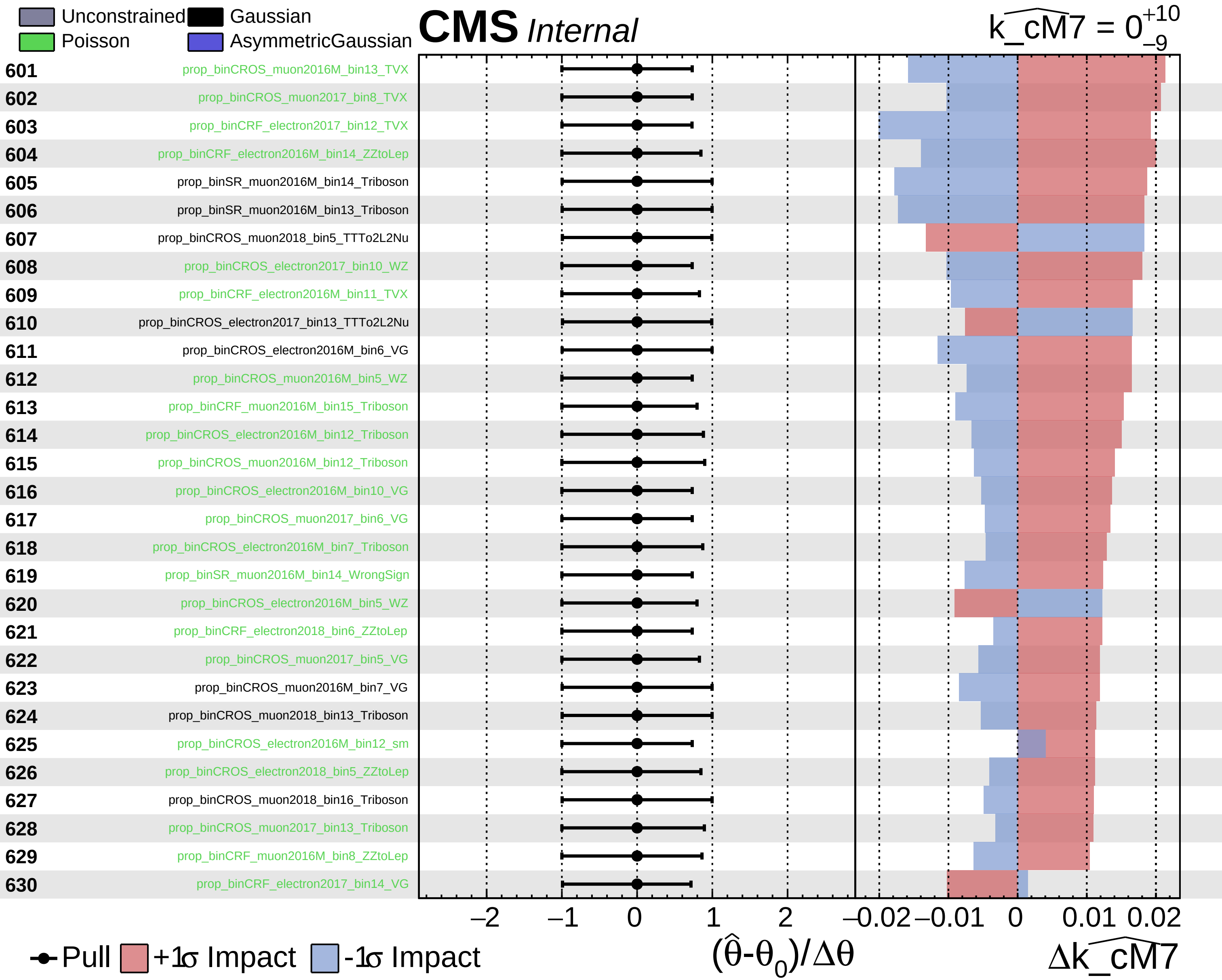


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
511 prop_binSR_electron2017_bin6_WpWpJJ_QCD
512 prop_binCROS_electron2018_bin5_TTTo2L2Nu
513 prop_binCRF_electron2017_bin9
514 lep_2016M
515 prop_binCROS_muon2017_bin8_TTTo2L2Nu
516 prop_binCRF_electron2017_bin11
517 prop_binCRF_muon2016M_bin15_WrongSign
518 prop_binCRF_electron2016M_bin14_Triboson
519 prop_binCROS_muon2017_bin8_VG
520 prop_binCRTT_muon2017_bin7
521 prop_binCRTT_electron2016M_bin9
522 prop_binCRF_electron2016M_bin15_WrongSign
523 prop_binCRF_electron2018_bin8
524 prop_binCROS_electron2018_bin6_Fake
525 prop_binCROS_muon2018_bin16_WZ
526 prop_binCROS_electron2017_bin13_Fake
527 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
528 prop_binCROS_electron2017_bin9_WZ
529 prop_binCROS_electron2017_bin9_TTTo2L2Nu
530 prop_binSR_muon2016M_bin14_ZZtoLep
531 prop_binCROS_muon2017_bin6_TTTo2L2Nu
532 prop_binCRTT_muon2018_bin10
533 prop_binCROS_electron2016M_bin6_TTTo2L2Nu
534 prop_binCRF_muon2016M_bin11_TVX
535 prop_binCROS_muon2018_bin13_VG
536 prop_binCRF_muon2016M_bin15_TVX
537 prop_binCROS_muon2017_bin13_TVX
538 lep_2018
539 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
540 prop_binCRTT_electron2016M_bin8
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$

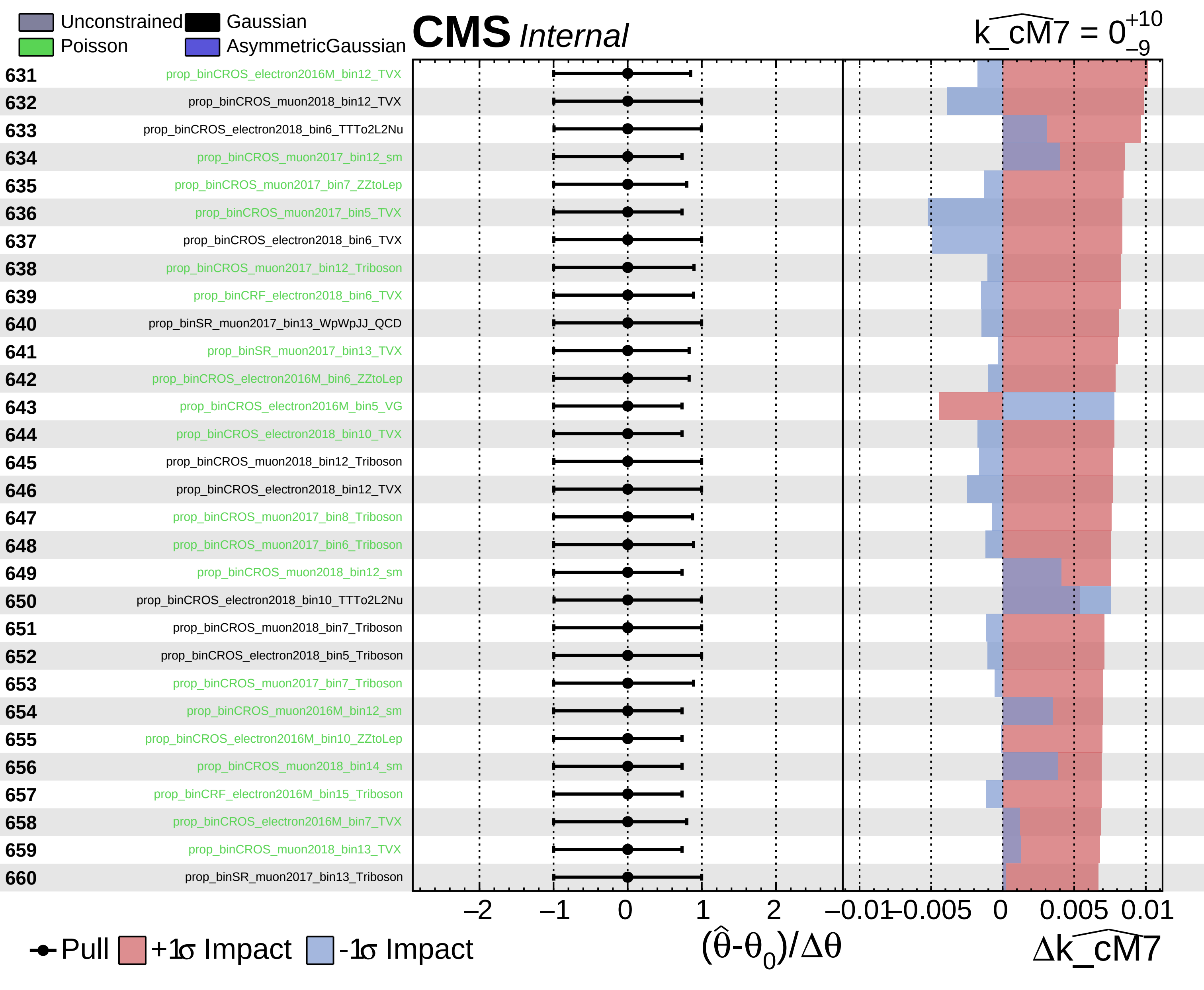


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
541 prop_binCROS_muon2016M_bin13_TTTo2L2Nu
542 prop_binCRTT_electron2018_bin3
543 prop_binCRF_muon2016M_bin8_WpWpJJ_QCD
544 prop_binCROS_electron2017_bin9_VG
545 prop_binSR_electron2017_bin6_WrongSign
546 prop_binCROS_electron2017_bin10_TTTo2L2Nu
547 prop_binCROS_muon2016M_bin5_TTTo2L2Nu
548 prop_binSR_muon2016M_bin13_TVX
549 prop_binCRF_electron2018_bin6_sm
550 lep_2017
551 prop_binCROS_muon2018_bin12_TTTo2L2Nu
552 prop_binSR_electron2017_bin13_WpWpJJ_QCD
553 prop_binCROS_electron2017_bin16
554 prop_binCRTT_muon2017_bin6
555 prop_binSR_electron2017_bin13_WrongSign
556 prop_binCRF_muon2016M_bin15_WpWpJJ_QCD
557 prop_binSR_electron2017_bin13_TVX
558 prop_binCROS_muon2018_bin14_TTTo2L2Nu
559 prop_binSR_electron2017_bin6_Triboson
560 prop_binCROS_muon2017_bin10_VG
561 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
562 prop_binCROS_electron2018_bin5_TVX
563 prop_binCRF_electron2016M_bin11_ZZtoLep
564 prop_binCROS_muon2017_bin12_TVX
565 QCDScale_ZZtoLep
566 prop_binCROS_electron2016M_bin14_WZ
567 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
568 prop_binCROS_electron2017_bin9_TVX
569 prop_binCRF_electron2016M_bin15_TVX
570 prop_binCROS_muon2018_bin7_VG
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$

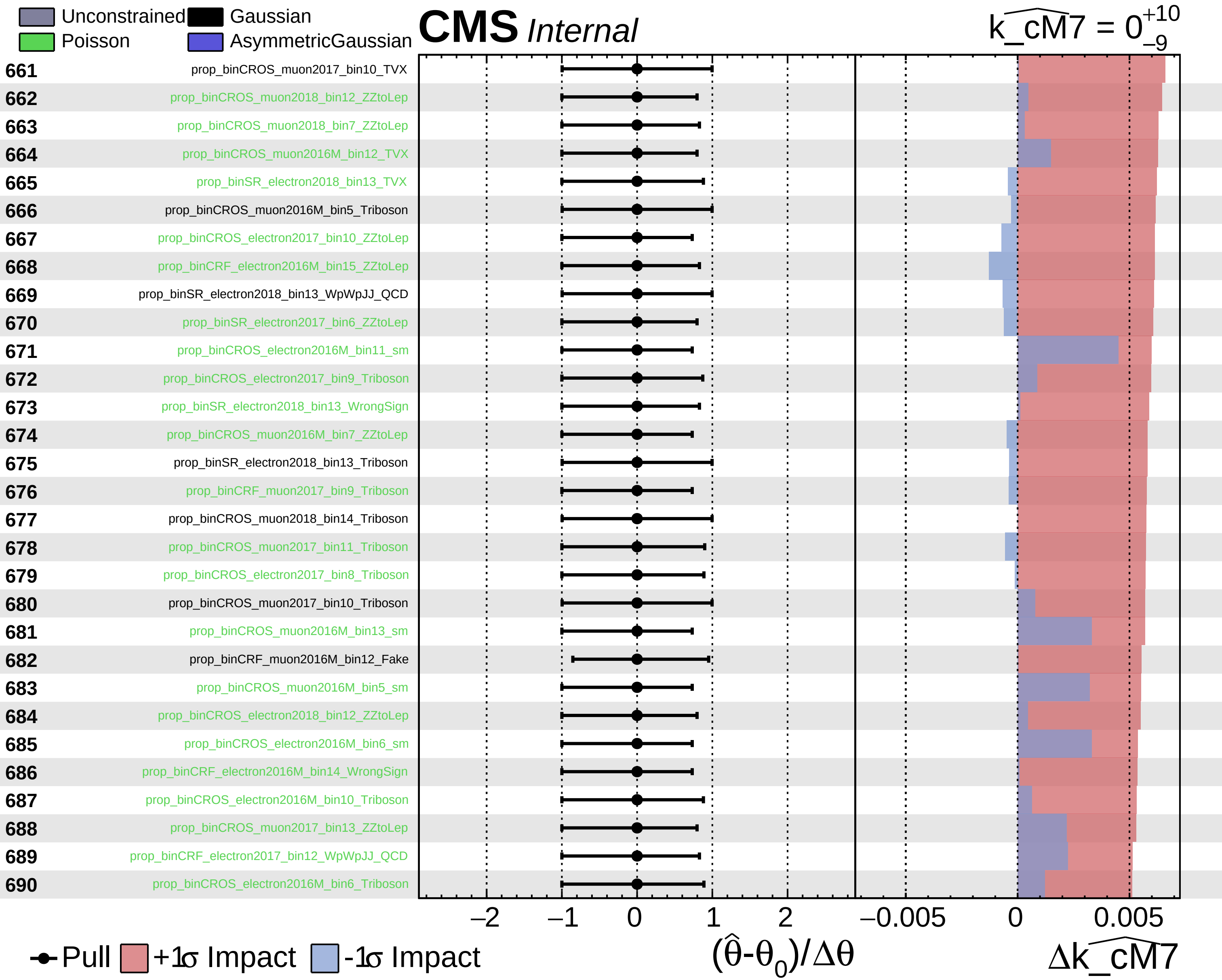

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
571 prop_binCRTT_muon2016M_bin13
572 prop_binCROS_muon2018_bin16_TVX
573 prop_binCROS_electron2016M_bin7_VG
574 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
575 prop_binCRF_muon2016M_bin11_WpWpJJ_QCD
576 prop_binCROS_electron2017_bin8_TTTo2L2Nu
577 prop_binSR_electron2017_bin13_Triboson
578 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
579 prop_binCROS_electron2018_bin12_TTTo2L2Nu
580 prop_binCROS_electron2016M_bin10_TTTo2L2Nu
581 prop_binCROS_electron2016M_bin6_WZ
582 prop_binSR_muon2016M_bin14_TVX
583 prop_binCROS_electron2018_bin8_TTTo2L2Nu
584 prop_binCROS_muon2016M_bin13_WZ
585 prop_binCROS_muon2017_bin11_VG
586 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
587 prop_binCROS_muon2017_bin7_VG
588 prop_binCROS_electron2017_bin8_WZ
589 prop_binCROS_muon2017_bin5_TTTo2L2Nu
590 prop_binCROS_muon2016M_bin5_VG
591 prop_binCROS_muon2017_bin7_TVX
592 prop_binCROS_muon2016M_bin5_TVX
593 prop_binCRF_muon2016M_bin11_Triboson
594 prop_binSR_muon2017_bin13_ZZtoLep
595 prop_binCROS_muon2017_bin15_Fake
596 prop_binCRF_muon2016M_bin8_Triboson
597 prop_binCROS_muon2017_bin15_TTTo2L2Nu
598 prop_binCROS_muon2018_bin16_sm
599 prop_binSR_electron2018_bin13_TTTo2L2Nu
600 prop_binCRF_electron2016M_bin15_WpWpJJ_QCD
−2
−1
0
1
2
−0.04
−0.02
0
0.02
0.04
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



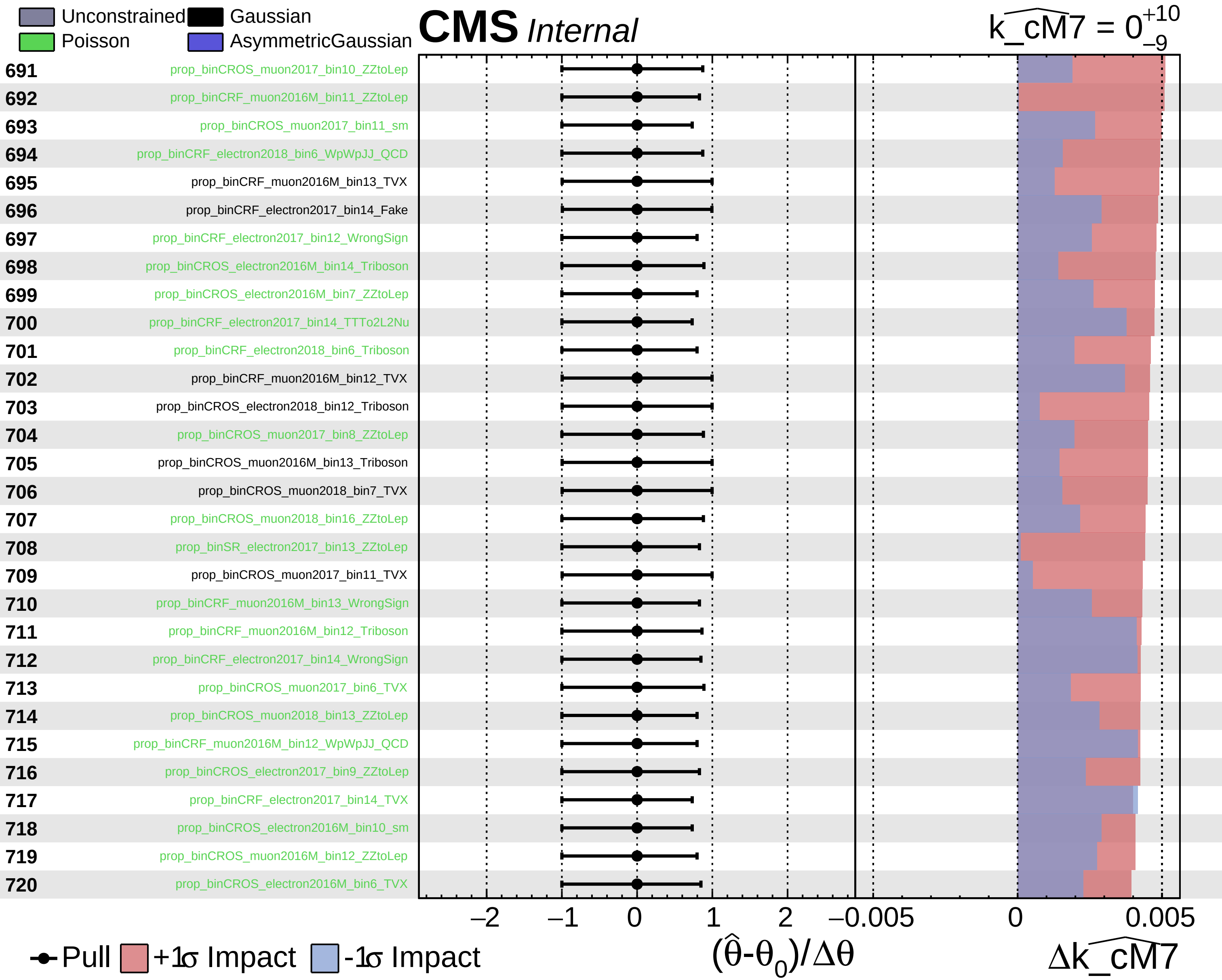
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0^{+10}_{-9}
601 prop_binCROS_muon2016M_bin13_TVX
602 prop_binCROS_muon2017_bin8_TVX
603 prop_binCRF_electron2017_bin12_TVX
604 prop_binCRF_electron2016M_bin14_ZZtoLep
605 prop_binSR_muon2016M_bin14_Triboson
606 prop_binSR_muon2016M_bin13_Triboson
607 prop_binCROS_muon2018_bin5_TTTo2L2Nu
608 prop_binCROS_electron2017_bin10_WZ
609 prop_binCRF_electron2016M_bin11_TVX
610 prop_binCROS_electron2017_bin13_TTTo2L2Nu
611 prop_binCROS_electron2016M_bin6_VG
612 prop_binCROS_muon2016M_bin5_WZ
613 prop_binCRF_muon2016M_bin15_Triboson
614 prop_binCROS_electron2016M_bin12_Triboson
615 prop_binCROS_muon2016M_bin12_Triboson
616 prop_binCROS_electron2016M_bin10_VG
617 prop_binCROS_muon2017_bin6_VG
618 prop_binCROS_electron2016M_bin7_Triboson
619 prop_binSR_muon2016M_bin14_WrongSign
620 prop_binCROS_electron2016M_bin5_WZ
621 prop_binCRF_electron2018_bin6_ZZtoLep
622 prop_binCROS_muon2017_bin5_VG
623 prop_binCROS_muon2016M_bin7_VG
624 prop_binCROS_muon2018_bin13_Triboson
625 prop_binCROS_electron2016M_bin12_sm
626 prop_binCROS_electron2018_bin5_ZZtoLep
627 prop_binCROS_muon2018_bin16_Triboson
628 prop_binCROS_muon2017_bin13_Triboson
629 prop_binCRF_muon2016M_bin8_ZZtoLep
630 prop_binCRF_electron2017_bin14_VG
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



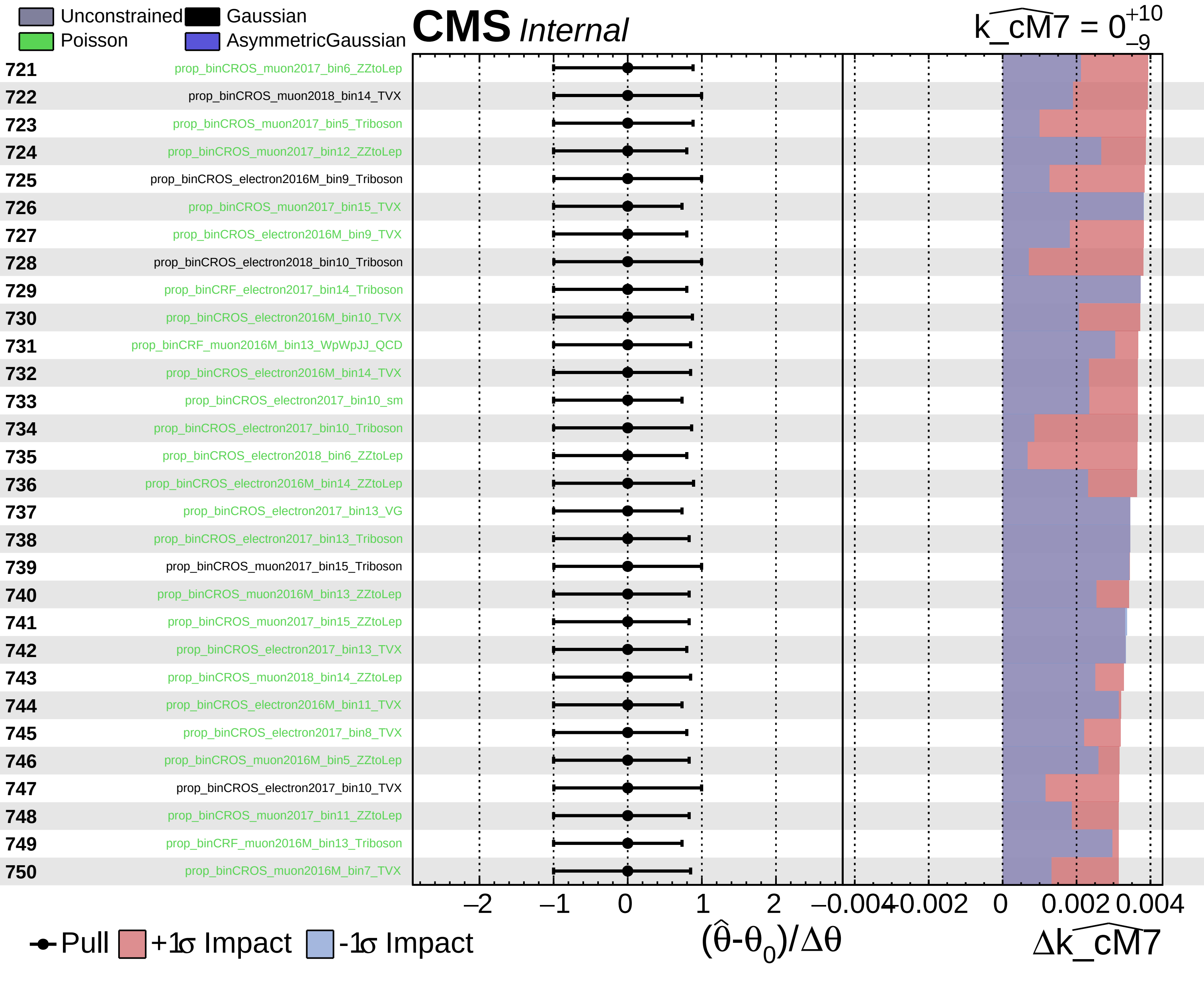
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +10 −9
631 prop_binCROS_electron2016M_bin12_TVX
632 prop_binCROS_muon2018_bin12_TVX
633 prop_binCROS_electron2018_bin6_TTTo2L2Nu
634 prop_binCROS_muon2017_bin12_sm
635 prop_binCROS_muon2017_bin7_ZZtoLep
636 prop_binCROS_muon2017_bin5_TVX
637 prop_binCROS_electron2018_bin6_TVX
638 prop_binCROS_muon2017_bin12_Triboson
639 prop_binCRF_electron2018_bin6_TVX
640 prop_binSR_muon2017_bin13_WpWpJJ_QCD
641 prop_binSR_muon2017_bin13_TVX
642 prop_binCROS_electron2016M_bin6_ZZtoLep
643 prop_binCROS_electron2016M_bin5_VG
644 prop_binCROS_electron2018_bin10_TVX
645 prop_binCROS_muon2018_bin12_Triboson
646 prop_binCROS_electron2018_bin12_TVX
647 prop_binCROS_muon2017_bin8_Triboson
648 prop_binCROS_muon2017_bin6_Triboson
649 prop_binCROS_muon2018_bin12_sm
650 prop_binCROS_electron2018_bin10_TTTo2L2Nu
651 prop_binCROS_muon2018_bin7_Triboson
652 prop_binCROS_electron2018_bin5_Triboson
653 prop_binCROS_muon2017_bin7_Triboson
654 prop_binCROS_muon2016M_bin12_sm
655 prop_binCROS_electron2016M_bin10_ZZtoLep
656 prop_binCROS_muon2018_bin14_sm
657 prop_binCRF_electron2016M_bin15_Triboson
658 prop_binCROS_electron2016M_bin7_TVX
659 prop_binCROS_muon2018_bin13_TVX
660 prop_binSR_muon2017_bin13_Triboson
−2
−1
0
1
2
−0.01
−0.005
0
0.005
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



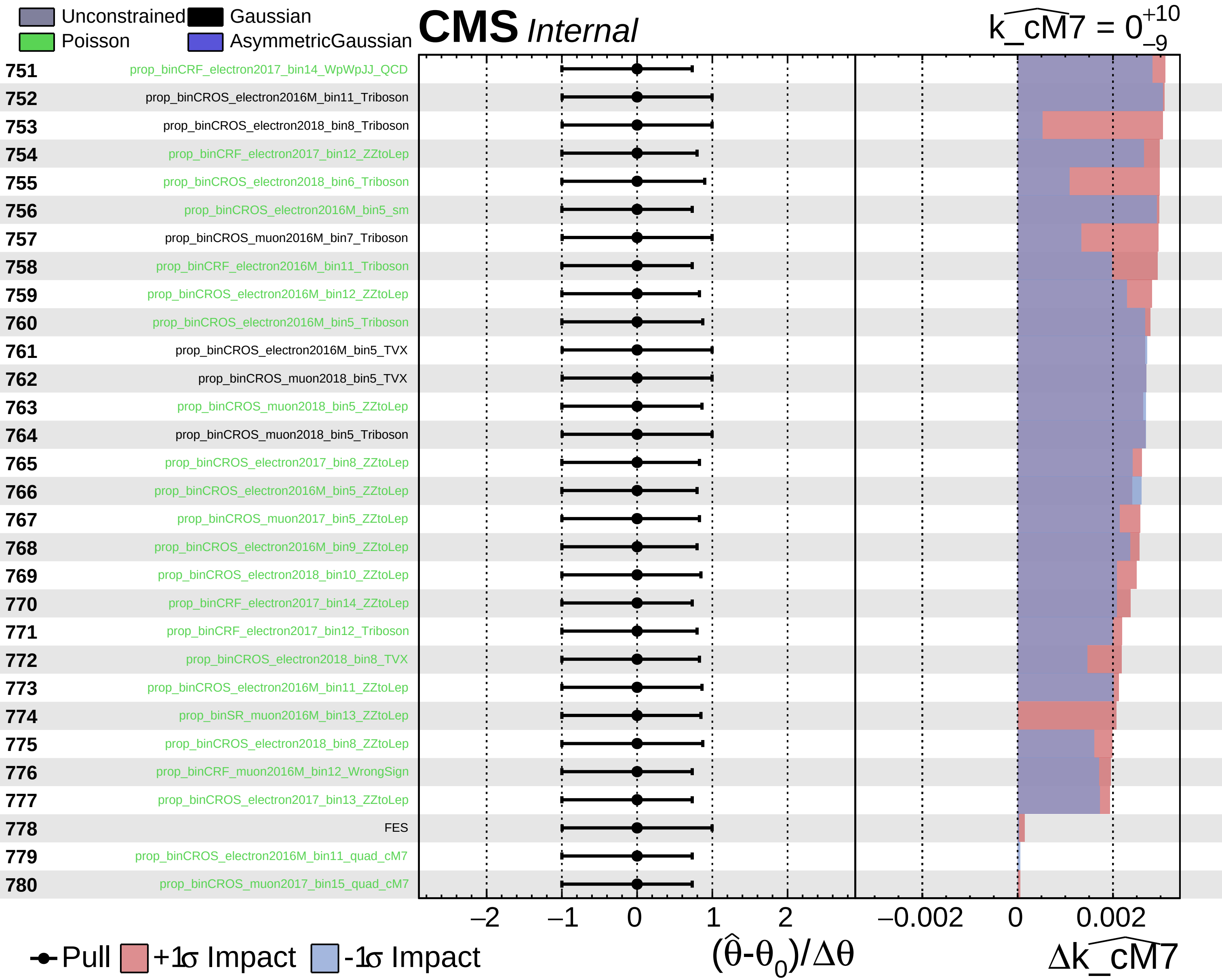
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
661 prop_binCROS_muon2017_bin10_TVX
662 prop_binCROS_muon2018_bin12_ZZtoLep
663 prop_binCROS_muon2018_bin7_ZZtoLep
664 prop_binCROS_muon2016M_bin12_TVX
665 prop_binSR_electron2018_bin13_TVX
666 prop_binCROS_muon2016M_bin5_Triboson
667 prop_binCROS_electron2017_bin10_ZZtoLep
668 prop_binCRF_electron2016M_bin15_ZZtoLep
669 prop_binSR_electron2018_bin13_WpWpJJ_QCD
670 prop_binSR_electron2017_bin6_ZZtoLep
671 prop_binCROS_electron2016M_bin11_sm
672 prop_binCROS_electron2017_bin9_Triboson
673 prop_binSR_electron2018_bin13_WrongSign
674 prop_binCROS_muon2016M_bin7_ZZtoLep
675 prop_binSR_electron2018_bin13_Triboson
676 prop_binCRF_muon2017_bin9_Triboson
677 prop_binCROS_muon2018_bin14_Triboson
678 prop_binCROS_muon2017_bin11_Triboson
679 prop_binCROS_electron2017_bin8_Triboson
680 prop_binCROS_muon2017_bin10_Triboson
681 prop_binCROS_muon2016M_bin13_sm
682 prop_binCRF_muon2016M_bin12_Fake
683 prop_binCROS_muon2016M_bin5_sm
684 prop_binCROS_electron2018_bin12_ZZtoLep
685 prop_binCROS_electron2016M_bin6_sm
686 prop_binCRF_electron2016M_bin14_WrongSign
687 prop_binCROS_electron2016M_bin10_Triboson
688 prop_binCROS_muon2017_bin13_ZZtoLep
689 prop_binCRF_electron2017_bin12_WpWpJJ_QCD
690 prop_binCROS_electron2016M_bin6_Triboson
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



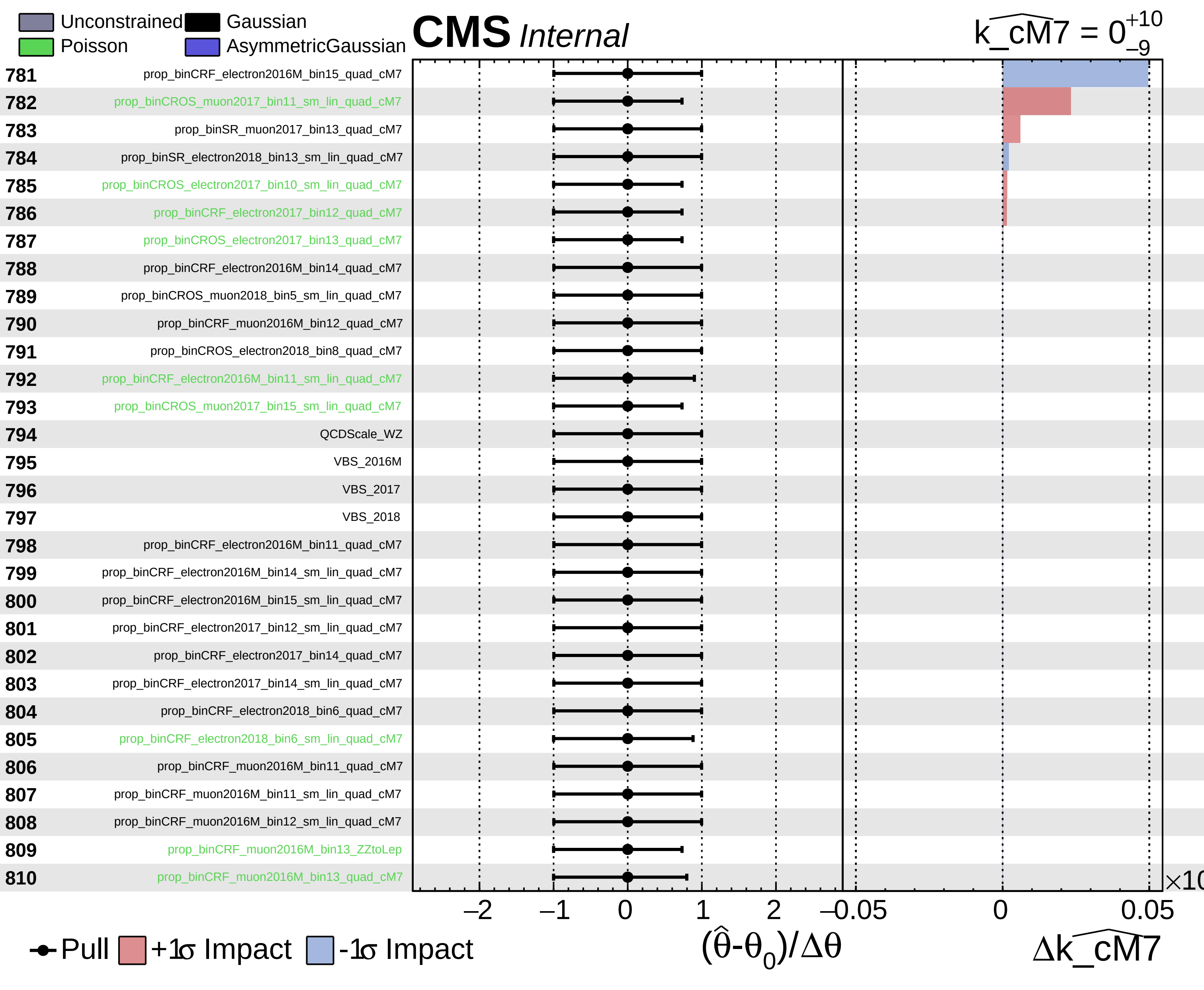
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
691 prop_binCROS_muon2017_bin10_ZZtoLep
692 prop_binCRF_muon2016M_bin11_ZZtoLep
693 prop_binCROS_muon2017_bin11_sm
694 prop_binCRF_electron2018_bin6_WpWpJJ_QCD
695 prop_binCRF_muon2016M_bin13_TVX
696 prop_binCRF_electron2017_bin14_Fake
697 prop_binCRF_electron2017_bin12_WrongSign
698 prop_binCROS_electron2016M_bin14_Triboson
699 prop_binCROS_electron2016M_bin7_ZZtoLep
700 prop_binCRF_electron2017_bin14_TTTo2L2Nu
701 prop_binCRF_electron2018_bin6_Triboson
702 prop_binCRF_muon2016M_bin12_TVX
703 prop_binCROS_electron2018_bin12_Triboson
704 prop_binCROS_muon2017_bin8_ZZtoLep
705 prop_binCROS_muon2016M_bin13_Triboson
706 prop_binCROS_muon2018_bin7_TVX
707 prop_binCROS_muon2018_bin16_ZZtoLep
708 prop_binSR_electron2017_bin13_ZZtoLep
709 prop_binCROS_muon2017_bin11_TVX
710 prop_binCRF_muon2016M_bin13_WrongSign
711 prop_binCRF_muon2016M_bin12_Triboson
712 prop_binCRF_electron2017_bin14_WrongSign
713 prop_binCROS_muon2017_bin6_TVX
714 prop_binCROS_muon2018_bin13_ZZtoLep
715 prop_binCRF_muon2016M_bin12_WpWpJJ_QCD
716 prop_binCROS_electron2017_bin9_ZZtoLep
717 prop_binCRF_electron2017_bin14_TVX
718 prop_binCROS_electron2016M_bin10_sm
719 prop_binCROS_muon2016M_bin12_ZZtoLep
720 prop_binCROS_electron2016M_bin6_TVX
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



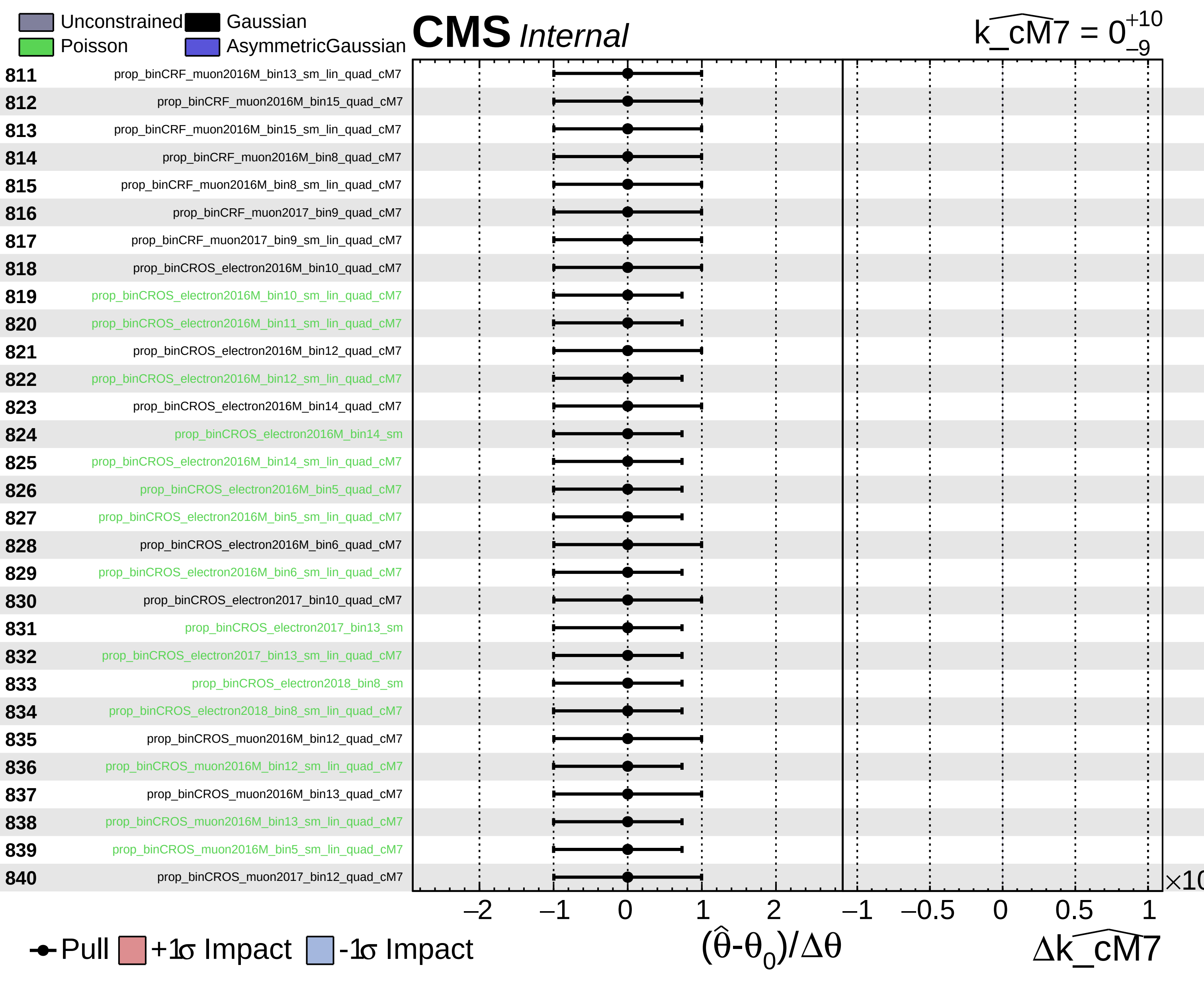
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +10 −9
721 prop_binCROS_muon2017_bin6_ZZtoLep
722 prop_binCROS_muon2018_bin14_TVX
723 prop_binCROS_muon2017_bin5_Triboson
724 prop_binCROS_muon2017_bin12_ZZtoLep
725 prop_binCROS_electron2016M_bin9_Triboson
726 prop_binCROS_muon2017_bin15_TVX
727 prop_binCROS_electron2016M_bin9_TVX
728 prop_binCROS_electron2018_bin10_Triboson
729 prop_binCRF_electron2017_bin14_Triboson
730 prop_binCROS_electron2016M_bin10_TVX
731 prop_binCRF_muon2016M_bin13_WpWpJJ_QCD
732 prop_binCROS_electron2016M_bin14_TVX
733 prop_binCROS_electron2017_bin10_sm
734 prop_binCROS_electron2017_bin10_Triboson
735 prop_binCROS_electron2018_bin6_ZZtoLep
736 prop_binCROS_electron2016M_bin14_ZZtoLep
737 prop_binCROS_electron2017_bin13_VG
738 prop_binCROS_electron2017_bin13_Triboson
739 prop_binCROS_muon2017_bin15_Triboson
740 prop_binCROS_muon2016M_bin13_ZZtoLep
741 prop_binCROS_muon2017_bin15_ZZtoLep
742 prop_binCROS_electron2017_bin13_TVX
743 prop_binCROS_muon2018_bin14_ZZtoLep
744 prop_binCROS_electron2016M_bin11_TVX
745 prop_binCROS_electron2017_bin8_TVX
746 prop_binCROS_muon2016M_bin5_ZZtoLep
747 prop_binCROS_electron2017_bin10_TVX
748 prop_binCROS_muon2017_bin11_ZZtoLep
749 prop_binCRF_muon2016M_bin13_Triboson
750 prop_binCROS_muon2016M_bin7_TVX
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cM7



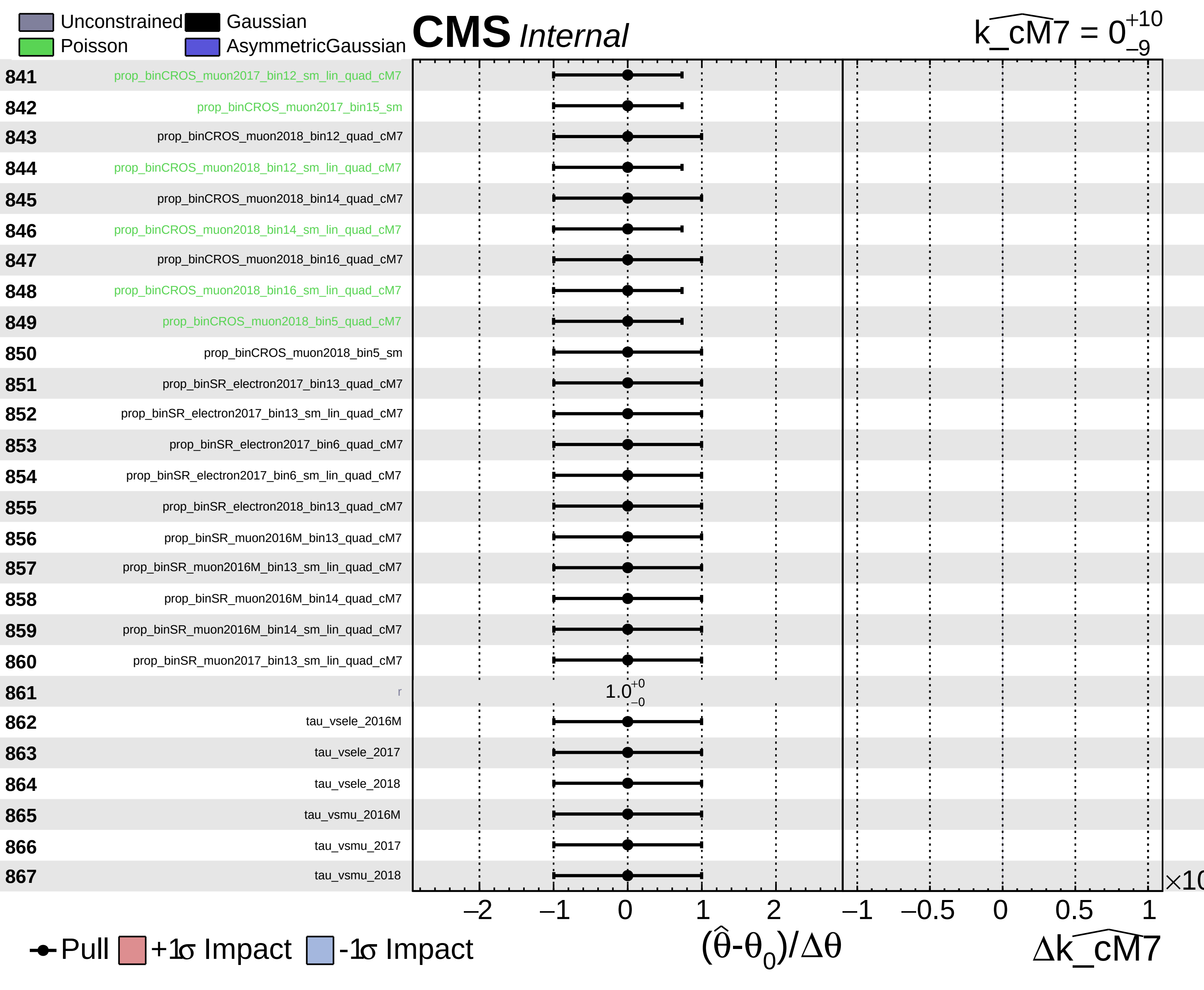
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +10 −9
751 prop_binCRF_electron2017_bin14_WpWpJJ_QCD
752 prop_binCROS_electron2016M_bin11_Triboson
753 prop_binCROS_electron2018_bin8_Triboson
754 prop_binCRF_electron2017_bin12_ZZtoLep
755 prop_binCROS_electron2018_bin6_Triboson
756 prop_binCROS_electron2016M_bin5_sm
757 prop_binCROS_muon2016M_bin7_Triboson
758 prop_binCRF_electron2016M_bin11_Triboson
759 prop_binCROS_electron2016M_bin12_ZZtoLep
760 prop_binCROS_electron2016M_bin5_Triboson
761 prop_binCROS_electron2016M_bin5_TVX
762 prop_binCROS_muon2018_bin5_TVX
763 prop_binCROS_muon2018_bin5_ZZtoLep
764 prop_binCROS_muon2018_bin5_Triboson
765 prop_binCROS_electron2017_bin8_ZZtoLep
766 prop_binCROS_electron2016M_bin5_ZZtoLep
767 prop_binCROS_muon2017_bin5_ZZtoLep
768 prop_binCROS_electron2016M_bin9_ZZtoLep
769 prop_binCROS_electron2018_bin10_ZZtoLep
770 prop_binCRF_electron2017_bin14_ZZtoLep
771 prop_binCRF_electron2017_bin12_Triboson
772 prop_binCROS_electron2018_bin8_TVX
773 prop_binCROS_electron2016M_bin11_ZZtoLep
774 prop_binSR_muon2016M_bin13_ZZtoLep
775 prop_binCROS_electron2018_bin8_ZZtoLep
776 prop_binCRF_muon2016M_bin12_WrongSign
777 prop_binCROS_electron2017_bin13_ZZtoLep
778 FES
779 prop_binCROS_electron2016M_bin11_quad_cM7
780 prop_binCROS_muon2017_bin15_quad_cM7
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
781 prop_binCRF_electron2016M_bin15_quad_cM7
782 prop_binCROS_muon2017_bin11_sm_lin_quad_cM7
783 prop_binSR_muon2017_bin13_quad_cM7
784 prop_binSR_electron2018_bin13_sm_lin_quad_cM7
785 prop_binCROS_electron2017_bin10_sm_lin_quad_cM7
786 prop_binCRF_electron2017_bin12_quad_cM7
787 prop_binCROS_electron2017_bin13_quad_cM7
788 prop_binCRF_electron2016M_bin14_quad_cM7
789 prop_binCROS_muon2018_bin5_sm_lin_quad_cM7
790 prop_binCRF_muon2016M_bin12_quad_cM7
791 prop_binCROS_electron2018_bin8_quad_cM7
792 prop_binCRF_electron2016M_bin11_sm_lin_quad_cM7
793 prop_binCROS_muon2017_bin15_sm_lin_quad_cM7
794 QCDScale_WZ
795 VBS_2016M
796 VBS_2017
797 VBS_2018
798 prop_binCRF_electron2016M_bin11_quad_cM7
799 prop_binCRF_electron2016M_bin14_sm_lin_quad_cM7
800 prop_binCRF_electron2016M_bin15_sm_lin_quad_cM7
801 prop_binCRF_electron2017_bin12_sm_lin_quad_cM7
802 prop_binCRF_electron2017_bin14_quad_cM7
803 prop_binCRF_electron2017_bin14_sm_lin_quad_cM7
804 prop_binCRF_electron2018_bin6_quad_cM7
805 prop_binCRF_electron2018_bin6_sm_lin_quad_cM7
806 prop_binCRF_muon2016M_bin11_quad_cM7
807 prop_binCRF_muon2016M_bin11_sm_lin_quad_cM7
808 prop_binCRF_muon2016M_bin12_sm_lin_quad_cM7
809 prop_binCRF_muon2016M_bin13_ZZtoLep
810 prop_binCRF_muon2016M_bin13_quad_cM7
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cM7 = 0 +10 −9
811 prop_binCRF_muon2016M_bin13_sm_lin_quad_cM7
812 prop_binCRF_muon2016M_bin15_quad_cM7
813 prop_binCRF_muon2016M_bin15_sm_lin_quad_cM7
814 prop_binCRF_muon2016M_bin8_quad_cM7
815 prop_binCRF_muon2016M_bin8_sm_lin_quad_cM7
816 prop_binCRF_muon2017_bin9_quad_cM7
817 prop_binCRF_muon2017_bin9_sm_lin_quad_cM7
818 prop_binCROS_electron2016M_bin10_quad_cM7
819 prop_binCROS_electron2016M_bin10_sm_lin_quad_cM7
820 prop_binCROS_electron2016M_bin11_sm_lin_quad_cM7
821 prop_binCROS_electron2016M_bin12_quad_cM7
822 prop_binCROS_electron2016M_bin12_sm_lin_quad_cM7
823 prop_binCROS_electron2016M_bin14_quad_cM7
824 prop_binCROS_electron2016M_bin14_sm
825 prop_binCROS_electron2016M_bin14_sm_lin_quad_cM7
826 prop_binCROS_electron2016M_bin5_quad_cM7
827 prop_binCROS_electron2016M_bin5_sm_lin_quad_cM7
828 prop_binCROS_electron2016M_bin6_quad_cM7
829 prop_binCROS_electron2016M_bin6_sm_lin_quad_cM7
830 prop_binCROS_electron2017_bin10_quad_cM7
831 prop_binCROS_electron2017_bin13_sm
832 prop_binCROS_electron2017_bin13_sm_lin_quad_cM7
833 prop_binCROS_electron2018_bin8_sm
834 prop_binCROS_electron2018_bin8_sm_lin_quad_cM7
835 prop_binCROS_muon2016M_bin12_quad_cM7
836 prop_binCROS_muon2016M_bin12_sm_lin_quad_cM7
837 prop_binCROS_muon2016M_bin13_quad_cM7
838 prop_binCROS_muon2016M_bin13_sm_lin_quad_cM7
839 prop_binCROS_muon2016M_bin5_sm_lin_quad_cM7
840 prop_binCROS_muon2017_bin12_quad_cM7
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cM7



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cM7} = 0^{+10}_{-9}$
841 prop_binCROS_muon2017_bin12_sm_lin_quad_cM7
842 prop_binCROS_muon2017_bin15_sm
843 prop_binCROS_muon2018_bin12_quad_cM7
844 prop_binCROS_muon2018_bin12_sm_lin_quad_cM7
845 prop_binCROS_muon2018_bin14_quad_cM7
846 prop_binCROS_muon2018_bin14_sm_lin_quad_cM7
847 prop_binCROS_muon2018_bin16_quad_cM7
848 prop_binCROS_muon2018_bin16_sm_lin_quad_cM7
849 prop_binCROS_muon2018_bin5_quad_cM7
850 prop_binCROS_muon2018_bin5_sm
851 prop_binSR_electron2017_bin13_quad_cM7
852 prop_binSR_electron2017_bin13_sm_lin_quad_cM7
853 prop_binSR_electron2017_bin6_quad_cM7
854 prop_binSR_electron2017_bin6_sm_lin_quad_cM7
855 prop_binSR_electron2018_bin13_quad_cM7
856 prop_binSR_muon2016M_bin13_quad_cM7
857 prop_binSR_muon2016M_bin13_sm_lin_quad_cM7
858 prop_binSR_muon2016M_bin14_quad_cM7
859 prop_binSR_muon2016M_bin14_sm_lin_quad_cM7
860 prop_binSR_muon2017_bin13_sm_lin_quad_cM7
861 r
1.0^{+0}_{-0}
862 tau_vsele_2016M
863 tau_vsele_2017
864 tau_vsele_2018
865 tau_vsmu_2016M
866 tau_vsmu_2017
867 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cM7}$